Appendix – R Programme Output

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Appendix I Logistic Regression Output (Full Dataset)

```
> start.time <- Sys.time()
>
> d<-read.csv("bank-additional-full-PREPROCESSED.csv") # load dataset
> d$duration<-NULL
> names(d)
 [1] "age"            "job"            "marital"        "education"
 [5] "default"        "housing"        "loan"           "contact"
 [9] "month"          "day_of_week"    "campaign"       "pdays"
[13] "previous"       "poutcome"       "emp.var.rate"   "cons.price.idx"
[17] "cons.conf.idx"  "euribor3m"      "nr.employed"    "y"
>
>
> set.seed(74020+82860+74011+71533)   # Use student id to generate sample
> n = max(row(d))
> id = sample(1:n, size = n*9/10)  # generate random row index for d_train
> d_train = d[id,]                        # training dataset
> d_test = d[-id,]                        # testing dataset
> dim(d_train)
[1] 37069    20
> dim(d_test)
[1] 4119   20
>
> lreg<-glm(d_train$y ~ ., data = d_train, binomial) # save logstic reg
> summary(lreg)

Call:
glm(formula = d_train$y ~ ., family = binomial, data = d_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0450  -0.3934  -0.3158  -0.2607   2.9410

Coefficients: (1 not defined because of singularities)
                              Estimate Std. Error z value Pr(>|z|)
(Intercept)                  -2.180e+02  3.519e+01  -6.194 5.87e-10 ***
age                           1.439e-04  2.241e-03   0.064 0.948815
jobblue-collar               -1.576e-01  7.241e-02  -2.177 0.029457 *
jobentrepreneur              -4.377e-02  1.124e-01  -0.389 0.697017
jobhousemaid                 -7.391e-02  1.343e-01  -0.550 0.582073
jobmanagement                -5.597e-02  7.908e-02  -0.708 0.479059
jobretired                    2.269e-01  9.996e-02   2.270 0.023203 *
jobself-employed             -4.318e-02  1.069e-01  -0.404 0.686404
jobservices                  -1.244e-01  7.924e-02  -1.570 0.116322
jobstudent                    1.946e-01  1.053e-01   1.848 0.064615 .
jobtechnician                -1.997e-02  6.555e-02  -0.305 0.760567
jobunemployed                -5.968e-02  1.180e-01  -0.506 0.612970
jobunknown                   -1.839e-01  2.222e-01  -0.828 0.407764
maritalmarried                5.104e-02  6.305e-02   0.809 0.418268
maritalsingle                 8.168e-02  7.197e-02   1.135 0.256444
maritalunknown                4.241e-01  3.765e-01   1.126 0.259962
educationbasic.6y             1.412e-01  1.099e-01   1.285 0.198899
educationbasic.9y             2.350e-02  8.743e-02   0.269 0.788122
educationhigh.school          7.844e-02  8.480e-02   0.925 0.354959
educationilliterate           8.841e-01  6.922e-01   1.277 0.201488
educationprofessional.course  8.132e-02  9.349e-02   0.870 0.384362
educationuniversity.degree    1.397e-01  8.489e-02   1.646 0.099768 .
educationunknown              1.858e-01  1.108e-01   1.678 0.093364 .
defaultunknown               -2.373e-01  6.068e-02  -3.912 9.17e-05 ***
```

```
defaultyes                  -8.633e+00  1.135e+02  -0.076 0.939364
housingunknown              -2.442e-02  1.226e-01  -0.199 0.842163
housingyes                  -2.410e-02  3.804e-02  -0.634 0.526390
loanunknown                         NA         NA      NA       NA
loanyes                     -5.724e-02  5.303e-02  -1.079 0.280465
contacttelephone            -7.362e-01  7.028e-02 -10.476  < 2e-16 ***
monthaug                     4.423e-01  1.132e-01   3.906 9.37e-05 ***
monthdec                     3.407e-01  1.982e-01   1.719 0.085630 .
monthjul                     6.846e-02  8.764e-02   0.781 0.434745
monthjun                    -6.652e-01  1.166e-01  -5.703 1.18e-08 ***
monthmar                     1.446e+00  1.365e-01  10.592  < 2e-16 ***
monthmay                    -4.191e-01  7.587e-02  -5.524 3.31e-08 ***
monthnov                    -4.832e-01  1.109e-01  -4.357 1.32e-05 ***
monthoct                     4.023e-02  1.427e-01   0.282 0.777987
monthsep                     2.233e-01  1.672e-01   1.336 0.181680
day_of_weekmon              -2.180e-01  6.118e-02  -3.563 0.000367 ***
day_of_weekthu               7.724e-02  5.872e-02   1.315 0.188376
day_of_weektue               7.060e-02  6.038e-02   1.169 0.242332
day_of_weekwed               1.174e-01  6.042e-02   1.944 0.051938 .
campaign                    -4.036e-02  9.626e-03  -4.193 2.75e-05 ***
pdays                       -1.057e-03  2.110e-04  -5.008 5.49e-07 ***
previous                    -7.755e-02  5.836e-02  -1.329 0.183910
poutcomenonexistent          4.226e-01  9.045e-02   4.673 2.97e-06 ***
poutcomesuccess              7.743e-01  2.064e-01   3.751 0.000176 ***
emp.var.rate                -1.496e+00  1.298e-01 -11.522  < 2e-16 ***
cons.price.idx               2.017e+00  2.316e-01   8.707  < 2e-16 ***
cons.conf.idx                2.615e-02  7.390e-03   3.539 0.000402 ***
euribor3m                    2.740e-01  1.213e-01   2.260 0.023841 *
nr.employed                  5.433e-03  2.872e-03   1.892 0.058497 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 26144  on 37068  degrees of freedom
Residual deviance: 20442  on 37017  degrees of freedom
AIC: 20546

Number of Fisher Scoring iterations: 10

> lreg<-step(lreg)
 # perform stepwise selection
Start:  AIC=20545.86
d_train$y ~ age + job + marital + education + default + housing +
    loan + contact + month + day_of_week + campaign + pdays +
    previous + poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    euribor3m + nr.employed

                Df Deviance   AIC
- education      7    20449 20539
- marital        3    20444 20542
- age            1    20442 20544
- housing        1    20442 20544
- loan           1    20443 20545
- previous       1    20444 20546
<none>                20442 20546
- nr.employed    1    20445 20547
- job           11    20466 20548
- euribor3m      1    20447 20549
- cons.conf.idx  1    20454 20556
- default        2    20458 20558
```

```
- campaign        1    20462 20564
- pdays           1    20467 20569
- poutcome        2    20472 20572
- day_of_week     4    20483 20579
- cons.price.idx  1    20517 20619
- contact         1    20562 20664
- emp.var.rate    1    20573 20675
- month           9    20810 20896

Step:  AIC=20539.04
d_train$y ~ age + job + marital + default + housing + loan +
    contact + month + day_of_week + campaign + pdays + previous +
    poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    euribor3m + nr.employed

                 Df Deviance   AIC
- marital         3    20452 20536
- age             1    20449 20537
- housing         1    20449 20537
- loan            1    20450 20538
- previous        1    20451 20539
<none>                20449 20539
- nr.employed     1    20453 20541
- euribor3m       1    20454 20542
- job            11    20479 20547
- cons.conf.idx   1    20462 20550
- default         2    20466 20552
- campaign        1    20468 20556
- pdays           1    20474 20562
- poutcome        2    20480 20566
- day_of_week     4    20490 20572
- cons.price.idx  1    20524 20612
- contact         1    20570 20658
- emp.var.rate    1    20581 20669
- month           9    20822 20894

Step:  AIC=20535.55
d_train$y ~ age + job + default + housing + loan + contact +
    month + day_of_week + campaign + pdays + previous + poutcome +
    emp.var.rate + cons.price.idx + cons.conf.idx + euribor3m +
    nr.employed

                 Df Deviance   AIC
- age             1    20452 20534
- housing         1    20452 20534
- loan            1    20453 20535
- previous        1    20453 20535
<none>                20452 20536
- nr.employed     1    20455 20537
- euribor3m       1    20457 20539
- job            11    20484 20546
- cons.conf.idx   1    20464 20546
- default         2    20468 20548
- campaign        1    20471 20553
- pdays           1    20476 20558
- poutcome        2    20483 20563
- day_of_week     4    20492 20568
- cons.price.idx  1    20527 20609
- contact         1    20573 20655
- emp.var.rate    1    20583 20665
- month           9    20826 20892
```

```
Step:  AIC=20533.86
d_train$y ~ job + default + housing + loan + contact + month +
    day_of_week + campaign + pdays + previous + poutcome + emp.var.rate +
    cons.price.idx + cons.conf.idx + euribor3m + nr.employed

                Df Deviance   AIC
- housing        1    20452 20532
- loan           1    20453 20533
- previous       1    20454 20534
<none>                20452 20534
- nr.employed    1    20455 20535
- euribor3m      1    20457 20537
- job           11    20484 20544
- cons.conf.idx  1    20464 20544
- default        2    20469 20547
- campaign       1    20471 20551
- pdays          1    20476 20556
- poutcome       2    20483 20561
- day_of_week    4    20493 20567
- cons.price.idx 1    20527 20607
- contact        1    20573 20653
- emp.var.rate   1    20583 20663
- month          9    20826 20890

Step:  AIC=20532.27
d_train$y ~ job + default + loan + contact + month + day_of_week +
    campaign + pdays + previous + poutcome + emp.var.rate + cons.price.idx +
    cons.conf.idx + euribor3m + nr.employed

                Df Deviance   AIC
- loan           2    20454 20530
- previous       1    20454 20532
<none>                20452 20532
- nr.employed    1    20456 20534
- euribor3m      1    20458 20536
- job           11    20484 20542
- cons.conf.idx  1    20465 20543
- default        2    20470 20546
- campaign       1    20472 20550
- pdays          1    20477 20555
- poutcome       2    20483 20559
- day_of_week    4    20493 20565
- cons.price.idx 1    20528 20606
- contact        1    20573 20651
- emp.var.rate   1    20584 20662
- month          9    20827 20889

Step:  AIC=20529.55
d_train$y ~ job + default + contact + month + day_of_week + campaign +
    pdays + previous + poutcome + emp.var.rate + cons.price.idx +
    cons.conf.idx + euribor3m + nr.employed

                Df Deviance   AIC
- previous       1    20455 20529
<none>                20454 20530
- nr.employed    1    20457 20531
- euribor3m      1    20459 20533
- job           11    20486 20540
- cons.conf.idx  1    20466 20540
- default        2    20471 20543
```

```
- campaign        1    20473 20547
- pdays           1    20478 20552
- poutcome        2    20485 20557
- day_of_week     4    20494 20562
- cons.price.idx  1    20529 20603
- contact         1    20574 20648
- emp.var.rate    1    20586 20660
- month           9    20828 20886

Step:  AIC=20529.26
d_train$y ~ job + default + contact + month + day_of_week + campaign +
    pdays + poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    euribor3m + nr.employed

                 Df Deviance   AIC
<none>                20455 20529
- nr.employed     1    20459 20531
- euribor3m       1    20460 20532
- job            11    20487 20539
- cons.conf.idx   1    20468 20540
- default         2    20473 20543
- campaign        1    20475 20547
- pdays           1    20478 20550
- day_of_week     4    20496 20562
- cons.price.idx  1    20530 20602
- poutcome        2    20551 20621
- contact         1    20575 20647
- emp.var.rate    1    20587 20659
- month           9    20830 20886
> summary(lreg)

Call:
glm(formula = d_train$y ~ job + default + contact + month + day_of_week +
    campaign + pdays + poutcome + emp.var.rate + cons.price.idx +
    cons.conf.idx + euribor3m + nr.employed, family = binomial,
    data = d_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0435  -0.3923  -0.3168  -0.2617   2.9872

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)        -2.163e+02  3.512e+01  -6.158 7.34e-10 ***
jobblue-collar     -2.135e-01  5.959e-02  -3.583 0.000340 ***
jobentrepreneur    -5.820e-02  1.109e-01  -0.525 0.599630
jobhousemaid       -1.348e-01  1.281e-01  -1.052 0.292584
jobmanagement      -5.023e-02  7.732e-02  -0.650 0.515898
jobretired          1.674e-01  7.957e-02   2.104 0.035339 *
jobself-employed   -4.314e-02  1.060e-01  -0.407 0.683991
jobservices        -1.562e-01  7.537e-02  -2.073 0.038169 *
jobstudent          1.958e-01  9.636e-02   2.032 0.042171 *
jobtechnician      -3.495e-02  5.832e-02  -0.599 0.548911
jobunemployed      -9.465e-02  1.164e-01  -0.813 0.416264
jobunknown         -1.711e-01  2.186e-01  -0.783 0.433828
defaultunknown     -2.430e-01  5.978e-02  -4.066 4.79e-05 ***
defaultyes         -8.634e+00  1.134e+02  -0.076 0.939326
contacttelephone   -7.318e-01  7.007e-02 -10.444  < 2e-16 ***
monthaug            4.428e-01  1.129e-01   3.922 8.78e-05 ***
monthdec            3.343e-01  1.978e-01   1.690 0.091001 .
monthjul            7.115e-02  8.736e-02   0.814 0.415419
```

```
monthjun            -6.575e-01  1.162e-01  -5.656 1.55e-08 ***
monthmar             1.447e+00  1.363e-01  10.618  < 2e-16 ***
monthmay            -4.246e-01  7.561e-02  -5.615 1.96e-08 ***
monthnov            -4.899e-01  1.107e-01  -4.424 9.68e-06 ***
monthoct             3.561e-02  1.426e-01   0.250 0.802726
monthsep             2.210e-01  1.670e-01   1.323 0.185901
day_of_weekmon      -2.190e-01  6.111e-02  -3.584 0.000338 ***
day_of_weekthu       7.854e-02  5.865e-02   1.339 0.180540
day_of_weektue       6.725e-02  6.033e-02   1.115 0.264983
day_of_weekwed       1.148e-01  6.038e-02   1.902 0.057192 .
campaign            -4.035e-02  9.617e-03  -4.196 2.72e-05 ***
pdays               -9.573e-04  1.979e-04  -4.838 1.31e-06 ***
poutcomenonexistent  5.149e-01  5.973e-02   8.621  < 2e-16 ***
poutcomesuccess      8.524e-01  1.990e-01   4.282 1.85e-05 ***
emp.var.rate        -1.494e+00  1.297e-01 -11.521  < 2e-16 ***
cons.price.idx       2.000e+00  2.309e-01   8.659  < 2e-16 ***
cons.conf.idx        2.628e-02  7.358e-03   3.571 0.000355 ***
euribor3m            2.758e-01  1.211e-01   2.277 0.022778 *
nr.employed          5.408e-03  2.866e-03   1.887 0.059175 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 26144  on 37068  degrees of freedom
Residual deviance: 20455  on 37032  degrees of freedom
AIC: 20529

Number of Fisher Scoring iterations: 10

> readline("Hit <Return> to continue:")
Hit <Return> to continue:
[1] ""
>
> c1 <- predict(lreg, d_train)                 # Prediction on Training Data S
et
> pred<-exp(c1)/(1+exp(c1))
> pred<-round((pred>0.5))
> table1<-table(pred,d_train$y)
> table1                                       # Display training data classifi
cation table

pred     0     1
   0 32394  3224
   1   488   963
> n<-sum(table1)                               # Number of data
> misclass_train<-(n-sum(diag(table1)))/n        # Compute training misclassif
ication rate
> misclass_train
[1] 0.1001376
>
>
> c2 <- predict(lreg, d_test)                    # Prediction on testing datas
et
> pred<-exp(c2)/(1+exp(c2))
> pred<-round((pred>0.5))
> table2<-table(pred,d_test$y)
> table2                                       # Display testing data classific
ation table

pred    0    1
```

```
   0 3615  341
   1   51  112
> n<-sum(table2)                                  # Number of data
> misclass_test<-(n-sum(diag(table2)))/n          # Compute testing misclassific
ation rate
> misclass_test
[1] 0.09516873
>
> end.time <- Sys.time()
> time.taken <- round(end.time - start.time,2)
> time.taken
Time difference of 2.61 mins
```

Appendix II Logistic Regression Output (10% Dataset)

```
> start.time <- Sys.time()
>
> d<-read.csv("bank-additional-PREPROCESSED.csv") # load dataset
> d$duration<-NULL
> names(d)
 [1] "age"            "job"            "marital"        "education"
 [5] "default"        "housing"        "loan"           "contact"
 [9] "month"          "day_of_week"    "campaign"       "pdays"
[13] "previous"       "poutcome"       "emp.var.rate"   "cons.price.idx"
[17] "cons.conf.idx"  "euribor3m"      "nr.employed"    "y"
>
>
> set.seed(74020+82860+74011+71533)   # Use student id to generate sample
> n = max(row(d))
> id = sample(1:n, size = n*9/10)  # generate random row index for d_train
> d_train = d[id,]                        # training dataset
> d_test = d[-id,]                        # testing dataset
> dim(d_train)
[1] 3707   20
> dim(d_test)
[1] 412  20
>
> lreg<-glm(d_train$y ~ ., data = d_train, binomial) # save logstic reg
> summary(lreg)

Call:
glm(formula = d_train$y ~ ., family = binomial, data = d_train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0825  -0.4036  -0.3228  -0.2559   2.7660

Coefficients: (1 not defined because of singularities)
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -2.062e+02  1.118e+02  -1.845 0.065068 .
age                         1.416e-02  7.210e-03   1.964 0.049566 *
jobblue-collar             -3.195e-01  2.359e-01  -1.354 0.175630
jobentrepreneur            -4.722e-01  4.024e-01  -1.173 0.240601
jobhousemaid               -1.998e-01  4.160e-01  -0.480 0.631039
jobmanagement              -4.406e-01  2.599e-01  -1.695 0.089987 .
jobretired                 -2.360e-01  3.176e-01  -0.743 0.457500
jobself-employed           -7.974e-01  3.888e-01  -2.051 0.040277 *
jobservices                -1.751e-01  2.485e-01  -0.705 0.481001
jobstudent                 -3.966e-01  3.863e-01  -1.027 0.304562
```

```
jobtechnician                 -7.939e-02  2.019e-01  -0.393 0.694175
jobunemployed                  1.458e-01  3.399e-01   0.429 0.667809
jobunknown                    -3.907e-01  7.013e-01  -0.557 0.577497
maritalmarried                 1.643e-01  2.127e-01   0.773 0.439807
maritalsingle                  3.141e-01  2.421e-01   1.297 0.194534
maritalunknown                 1.219e-01  1.172e+00   0.104 0.917110
educationbasic.6y              2.681e-01  3.544e-01   0.756 0.449353
educationbasic.9y              7.173e-02  2.884e-01   0.249 0.803610
educationhigh.school           1.229e-01  2.732e-01   0.450 0.652832
educationilliterate           -1.104e+01  3.247e+02  -0.034 0.972893
educationprofessional.course   2.805e-01  2.988e-01   0.939 0.347803
educationuniversity.degree     1.506e-01  2.749e-01   0.548 0.583818
educationunknown               3.113e-01  3.557e-01   0.875 0.381612
defaultunknown                -9.130e-02  1.891e-01  -0.483 0.629160
housingunknown                -6.900e-01  4.853e-01  -1.422 0.155031
housingyes                    -1.332e-01  1.229e-01  -1.084 0.278484
loanunknown                           NA         NA      NA       NA
loanyes                       -4.309e-02  1.650e-01  -0.261 0.794035
contacttelephone              -9.412e-01  2.446e-01  -3.848 0.000119 ***
monthaug                      -1.072e-02  3.751e-01  -0.029 0.977204
monthdec                       7.202e-01  6.328e-01   1.138 0.255081
monthjul                      -1.365e-02  3.091e-01  -0.044 0.964769
monthjun                      -2.636e-01  3.912e-01  -0.674 0.500317
monthmar                       1.906e+00  4.731e-01   4.029 5.61e-05 ***
monthmay                      -4.052e-01  2.636e-01  -1.537 0.124268
monthnov                      -5.488e-01  3.696e-01  -1.485 0.137638
monthoct                      -2.318e-01  4.775e-01  -0.486 0.627300
monthsep                       5.118e-02  5.397e-01   0.095 0.924446
day_of_weekmon                -4.004e-02  1.906e-01  -0.210 0.833629
day_of_weekthu                -1.012e-02  1.908e-01  -0.053 0.957687
day_of_weektue                -1.201e-01  1.981e-01  -0.606 0.544402
day_of_weekwed                 1.342e-01  1.964e-01   0.683 0.494312
campaign                      -6.540e-02  3.416e-02  -1.914 0.055579 .
pdays                         -7.982e-04  6.863e-04  -1.163 0.244806
previous                       1.546e-01  1.750e-01   0.884 0.376836
poutcomenonexistent            5.807e-01  2.875e-01   2.020 0.043392 *
poutcomesuccess                9.911e-01  6.812e-01   1.455 0.145703
emp.var.rate                  -1.166e+00  4.227e-01  -2.758 0.005816 **
cons.price.idx                 1.878e+00  7.381e-01   2.544 0.010951 *
cons.conf.idx                  5.167e-02  2.432e-02   2.125 0.033594 *
euribor3m                      6.411e-02  3.820e-01   0.168 0.866738
nr.employed                    5.856e-03  9.048e-03   0.647 0.517529
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2578.4  on 3706  degrees of freedom
Residual deviance: 2004.2  on 3656  degrees of freedom
AIC: 2106.2

Number of Fisher Scoring iterations: 11

> lreg<-step(lreg)
 # perform stepwise selection
Start:  AIC=2106.15
d_train$y ~ age + job + marital + education + default + housing +
    loan + contact + month + day_of_week + campaign + pdays +
    previous + poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    euribor3m + nr.employed
```

```
                  Df Deviance    AIC
- job             11   2013.9 2093.9
- education        7   2006.3 2094.3
- day_of_week      4   2005.9 2099.9
- marital          3   2006.0 2102.0
- euribor3m        1   2004.2 2104.2
- loan             1   2004.2 2104.2
- default          1   2004.4 2104.4
- nr.employed      1   2004.6 2104.6
- previous         1   2005.0 2105.0
- housing          1   2005.3 2105.3
- pdays            1   2005.5 2105.5
<none>                 2004.2 2106.2
- poutcome         2   2009.4 2107.4
- age              1   2008.0 2108.0
- campaign         1   2008.4 2108.4
- cons.conf.idx    1   2008.7 2108.7
- cons.price.idx   1   2010.6 2110.6
- emp.var.rate     1   2011.7 2111.7
- contact          1   2020.9 2120.9
- month            9   2048.5 2132.5

Step:  AIC=2093.94
d_train$y ~ age + marital + education + default + housing + loan +
    contact + month + day_of_week + campaign + pdays + previous +
    poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    euribor3m + nr.employed

                  Df Deviance    AIC
- education        7   2017.8 2083.8
- day_of_week      4   2015.7 2087.7
- marital          3   2016.0 2090.0
- euribor3m        1   2014.0 2092.0
- loan             1   2014.0 2092.0
- nr.employed      1   2014.2 2092.2
- default          1   2014.3 2092.3
- previous         1   2014.6 2092.6
- housing          1   2014.9 2092.9
- pdays            1   2015.3 2093.3
<none>                 2013.9 2093.9
- poutcome         2   2019.7 2095.7
- campaign         1   2018.2 2096.2
- cons.conf.idx    1   2018.2 2096.2
- age              1   2019.0 2097.0
- cons.price.idx   1   2019.7 2097.7
- emp.var.rate     1   2020.9 2098.9
- contact          1   2030.1 2108.1
- month            9   2058.3 2120.3

Step:  AIC=2083.76
d_train$y ~ age + marital + default + housing + loan + contact +
    month + day_of_week + campaign + pdays + previous + poutcome +
    emp.var.rate + cons.price.idx + cons.conf.idx + euribor3m +
    nr.employed

                  Df Deviance    AIC
- day_of_week      4   2019.4 2077.4
- marital          3   2019.8 2079.8
- loan             1   2017.8 2081.8
- euribor3m        1   2017.9 2081.9
- nr.employed      1   2017.9 2081.9
```

```
- default         1   2018.4 2082.4
- previous        1   2018.4 2082.4
- housing         1   2018.6 2082.6
- pdays           1   2019.1 2083.1
<none>                2017.8 2083.8
- poutcome        2   2023.3 2085.3
- campaign        1   2022.0 2085.9
- cons.conf.idx   1   2022.0 2086.0
- age             1   2022.2 2086.2
- cons.price.idx  1   2023.3 2087.3
- emp.var.rate    1   2024.7 2088.7
- contact         1   2034.5 2098.5
- month           9   2063.3 2111.3

Step:  AIC=2077.44
d_train$y ~ age + marital + default + housing + loan + contact +
    month + campaign + pdays + previous + poutcome + emp.var.rate +
    cons.price.idx + cons.conf.idx + euribor3m + nr.employed

                Df Deviance    AIC
- marital        3   2021.5 2073.6
- loan           1   2019.5 2075.5
- euribor3m      1   2019.5 2075.6
- nr.employed    1   2019.6 2075.6
- previous       1   2020.1 2076.1
- default        1   2020.1 2076.1
- housing        1   2020.3 2076.3
- pdays          1   2020.8 2076.8
<none>               2019.4 2077.4
- poutcome       2   2024.9 2078.9
- age            1   2023.7 2079.7
- cons.conf.idx  1   2023.8 2079.8
- campaign       1   2023.8 2079.8
- cons.price.idx 1   2025.1 2081.1
- emp.var.rate   1   2026.3 2082.3
- contact        1   2036.2 2092.2
- month          9   2064.3 2104.3

Step:  AIC=2073.55
d_train$y ~ age + default + housing + loan + contact + month +
    campaign + pdays + previous + poutcome + emp.var.rate + cons.price.idx +
    cons.conf.idx + euribor3m + nr.employed

                Df Deviance    AIC
- loan           1   2021.6 2071.6
- euribor3m      1   2021.6 2071.6
- nr.employed    1   2021.7 2071.7
- default        1   2022.2 2072.2
- previous       1   2022.2 2072.2
- housing        1   2022.4 2072.4
- pdays          1   2023.0 2072.9
<none>               2021.5 2073.6
- age            1   2024.1 2074.1
- poutcome       2   2027.1 2075.1
- campaign       1   2025.9 2075.9
- cons.conf.idx  1   2026.0 2076.0
- cons.price.idx 1   2027.2 2077.2
- emp.var.rate   1   2028.4 2078.4
- contact        1   2038.2 2088.2
- month          9   2068.0 2102.0
```

```
Step:  AIC=2071.62
d_train$y ~ age + default + housing + contact + month + campaign +
    pdays + previous + poutcome + emp.var.rate + cons.price.idx +
    cons.conf.idx + euribor3m + nr.employed

                Df Deviance    AIC
- euribor3m      1   2021.7 2069.7
- nr.employed    1   2021.8 2069.8
- previous       1   2022.3 2070.3
- default        1   2022.3 2070.3
- housing        2   2024.5 2070.5
- pdays          1   2023.0 2071.0
<none>               2021.6 2071.6
- age            1   2024.1 2072.1
- poutcome       2   2027.2 2073.2
- campaign       1   2025.9 2073.9
- cons.conf.idx  1   2026.1 2074.1
- cons.price.idx 1   2027.3 2075.3
- emp.var.rate   1   2028.5 2076.5
- contact        1   2038.5 2086.5
- month          9   2068.0 2100.0

Step:  AIC=2069.71
d_train$y ~ age + default + housing + contact + month + campaign +
    pdays + previous + poutcome + emp.var.rate + cons.price.idx +
    cons.conf.idx + nr.employed

                Df Deviance    AIC
- previous       1   2022.4 2068.4
- default        1   2022.4 2068.4
- nr.employed    1   2022.6 2068.6
- housing        2   2024.6 2068.6
- pdays          1   2023.1 2069.1
<none>               2021.7 2069.7
- age            1   2024.2 2070.2
- poutcome       2   2027.3 2071.3
- campaign       1   2026.1 2072.1
- emp.var.rate   1   2028.5 2074.5
- cons.price.idx 1   2029.3 2075.3
- cons.conf.idx  1   2032.5 2078.5
- contact        1   2038.6 2084.6
- month          9   2068.9 2098.9

Step:  AIC=2068.39
d_train$y ~ age + default + housing + contact + month + campaign +
    pdays + poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    nr.employed

                Df Deviance    AIC
- default        1   2023.1 2067.1
- housing        2   2025.1 2067.1
- nr.employed    1   2023.3 2067.3
<none>               2022.4 2068.4
- age            1   2025.0 2069.0
- pdays          1   2025.1 2069.1
- campaign       1   2026.8 2070.8
- poutcome       2   2029.1 2071.1
- emp.var.rate   1   2029.5 2073.5
- cons.price.idx 1   2030.6 2074.6
- cons.conf.idx  1   2033.3 2077.3
- contact        1   2039.9 2083.9
```

```
- month           9   2069.4 2097.4

Step:  AIC=2067.09
d_train$y ~ age + housing + contact + month + campaign + pdays +
    poutcome + emp.var.rate + cons.price.idx + cons.conf.idx +
    nr.employed

                 Df Deviance    AIC
- housing         2   2025.8 2065.8
- nr.employed     1   2024.0 2066.0
<none>                2023.1 2067.1
- age             1   2025.3 2067.3
- pdays           1   2025.8 2067.8
- campaign        1   2027.5 2069.5
- poutcome        2   2029.9 2069.9
- emp.var.rate    1   2030.2 2072.2
- cons.price.idx  1   2031.3 2073.3
- cons.conf.idx   1   2033.9 2075.9
- contact         1   2040.7 2082.7
- month           9   2070.4 2096.4

Step:  AIC=2065.85
d_train$y ~ age + contact + month + campaign + pdays + poutcome +
    emp.var.rate + cons.price.idx + cons.conf.idx + nr.employed

                 Df Deviance    AIC
- nr.employed     1   2026.8 2064.8
<none>                2025.8 2065.8
- age             1   2028.1 2066.1
- pdays           1   2028.5 2066.5
- campaign        1   2030.3 2068.3
- poutcome        2   2033.2 2069.2
- emp.var.rate    1   2033.1 2071.1
- cons.price.idx  1   2034.2 2072.2
- cons.conf.idx   1   2036.5 2074.5
- contact         1   2043.6 2081.6
- month           9   2072.5 2094.5

Step:  AIC=2064.82
d_train$y ~ age + contact + month + campaign + pdays + poutcome +
    emp.var.rate + cons.price.idx + cons.conf.idx

                 Df Deviance    AIC
<none>                2026.8 2064.8
- age             1   2029.1 2065.1
- pdays           1   2029.4 2065.4
- campaign        1   2031.3 2067.3
- poutcome        2   2034.3 2068.3
- cons.conf.idx   1   2036.5 2072.5
- contact         1   2043.8 2079.8
- month           9   2073.1 2093.1
- cons.price.idx  1   2086.6 2122.6
- emp.var.rate    1   2137.4 2173.4
> summary(lreg)

Call:
glm(formula = d_train$y ~ age + contact + month + campaign +
    pdays + poutcome + emp.var.rate + cons.price.idx + cons.conf.idx,
    family = binomial, data = d_train)

Deviance Residuals:
```

```
    Min       1Q   Median       3Q      Max
-2.0035  -0.3960  -0.3293  -0.2672   2.7507

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)         -1.184e+02  1.495e+01  -7.920 2.37e-15 ***
age                  7.711e-03  5.078e-03   1.519   0.1288
contacttelephone    -8.628e-01  2.198e-01  -3.926 8.65e-05 ***
monthaug            -8.709e-02  3.225e-01  -0.270   0.7871
monthdec             6.597e-01  5.756e-01   1.146   0.2517
monthjul             8.010e-02  2.965e-01   0.270   0.7870
monthjun             4.705e-02  2.770e-01   0.170   0.8651
monthmar             1.705e+00  3.919e-01   4.350 1.36e-05 ***
monthmay            -4.722e-01  2.397e-01  -1.970   0.0488 *
monthnov            -5.062e-01  2.951e-01  -1.715   0.0863 .
monthoct            -2.458e-01  3.906e-01  -0.629   0.5292
monthsep            -1.489e-01  3.944e-01  -0.378   0.7057
campaign            -6.758e-02  3.435e-02  -1.967   0.0492 *
pdays               -9.905e-04  6.115e-04  -1.620   0.1053
poutcomenonexistent  4.321e-01  1.887e-01   2.290   0.0220 *
poutcomesuccess      8.842e-01  6.183e-01   1.430   0.1527
emp.var.rate        -7.151e-01  6.555e-02 -10.909  < 2e-16 ***
cons.price.idx       1.270e+00  1.613e-01   7.872 3.50e-15 ***
cons.conf.idx        4.919e-02  1.591e-02   3.091   0.0020 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2578.4  on 3706  degrees of freedom
Residual deviance: 2026.8  on 3688  degrees of freedom
AIC: 2064.8

Number of Fisher Scoring iterations: 6

> readline("Hit <Return> to continue:")
Hit <Return> to continue:
[1] ""
>
> c1 <- predict(lreg, d_train)                  # Prediction on Training Data S
et
> pred<-exp(c1)/(1+exp(c1))
> pred<-round((pred>0.5))
> table1<-table(pred,d_train$y)
> table1                                        # Display training data classifi
cation table

pred    0    1
   0 3247  312
   1   50   98
> n<-sum(table1)                                # Number of data
> misclass_train<-(n-sum(diag(table1)))/n       # Compute training misclassif
ication rate
> misclass_train
[1] 0.09765309
>
>
> c2 <- predict(lreg, d_test)                     # Prediction on testing datas
et
> pred<-exp(c2)/(1+exp(c2))
> pred<-round((pred>0.5))
```

```
> table2<-table(pred,d_test$y)
> table2                                        # Display testing data classific
ation table

pred   0   1
   0 363  33
   1   8   8
> n<-sum(table2)                                # Number of data
> misclass_test<-(n-sum(diag(table2)))/n        # Compute testing misclassific
ation rate
> misclass_test
[1] 0.09951456
>
> end.time <- Sys.time()
> time.taken <- round(end.time - start.time,2)
> time.taken
Time difference of 56.59 secs
```

Appendix III Classification Tree Output (Full Dataset)

```
> start.time<-Sys.time()        # Measure running time
> library(rpart)                # load rpart library
> d<-read.csv("bank-additional-full-PREPROCESSED.csv") # load dataset
> d$duration<-NULL
> names(d)
 [1] "age"        "job"          "marital"      "education"     "default"

 [6] "housing     "loan"         "contact"      "month"         "day_of_week"

[11] "campaign"   "pdays"       "previous"      "poutcome"      "emp.var.rate"

[16] "cons.price.idx" "cons.conf.idx"  "euribor3m"      "nr.employed"    "y"

>
> set.seed(74020+82860+74011+71533)   # Use student id to generate sample
> n = max(row(d))
> id = sample(1:n, size = n*9/10)  # generate random row index for d_train
> d_train = d[id,]                      # training dataset
> d_test = d[-id,]                      # testing dataset
> dim(d_train)
[1] 37069    20
> dim(d_test)
[1] 4119   20
>
> ctree = rpart(d_train$y~.,data=d_train,
+ control=rpart.control(maxdepth=10),method="class")   # run the classificatio
n tree
```

```
> plot(ctree, asp=30, main="Business Success")     # Plot ctree # Please chang
e asp to suitable value
> text(ctree, use.n=T, cex = 0.6)                          # Add text   # Plea
se change cex to suitable value
> print(ctree)                                          # Write down the class
ification tree in words
n= 37069

node), split, n, loss, yval, (yprob)
      * denotes terminal node

1) root 37069 4187 0 (0.8870485 0.1129515)
  2) nr.employed>=5087.65 32583 2186 0 (0.9329098 0.0670902) *
  3) nr.employed< 5087.65 4486 2001 0 (0.5539456 0.4460544)
    6) pdays>=15.5 3391 1234 0 (0.6360955 0.3639045) *
    7) pdays< 15.5 1095  328 1 (0.2995434 0.7004566) *
>
> prob=predict(ctree, d_train)                          # Prediction on trainin
g data
> col=max.col(prob)                                     # Col is the col. no s.
t. pr is max
> table1 = table(col,d_train[,20])                      # Cross tabulation tabl
e for training dataset new_d
> table1

col     0     1
  1 32554  3420
  2   328   767
>
> n<-sum(table1)                                # Number of data
> misclass_d1<-(n-sum(diag(table1)))/n          # Compute d_train misclassificat
ion rate
> misclass_d1
[1] 0.1011087
>
> probD2=predict(ctree,d_test)                          # Prediction on testing
 data
> col2=max.col(probD2)                                  # Col is the col. no s.
t. pr is max
> table2 = table(col2,d_test[,20])                      # Cross tabulation tabl
e for training dataset d_test
> table2

col2    0    1
   1 3640  367
   2   26   86
>
> n<-sum(table2)                                # Number of data
> misclass_d2<-(n-sum(diag(table2)))/n          # Compute d_test misclassificati
on rate
> misclass_d2
[1] 0.09541151
>
> end.time <- Sys.time()
> time.taken <- round(end.time - start.time,2)
> time.taken
Time difference of 11.54 secs
```

Appendix IV Classification Tree Output (10% Dataset)

```
> start.time<-Sys.time()        # Measure running time
> library(rpart)                # load rpart library
> d<-read.csv("bank-additional-PREPROCESSED.csv") # load dataset
> d$duration<-NULL
> names(d)
 [1] "age"            "job"            "marital"        "education"
 [5] "default"        "housing"        "loan"           "contact"
 [9] "month"          "day_of_week"    "campaign"       "pdays"
[13] "previous"       "poutcome"       "emp.var.rate"   "cons.price.idx"
[17] "cons.conf.idx"  "euribor3m"      "nr.employed"    "y"
>
> set.seed(74020+82860+74011+71533)   # Use student id to generate sample
> n = max(row(d))
> id = sample(1:n, size = n*9/10)  # generate random row index for d_train
> d_train = d[id,]                      # training dataset
> d_test = d[-id,]                      # testing dataset
> dim(d_train)
[1] 3707   20
> dim(d_test)
[1] 412  20
>
```

```
> ctree = rpart(d_train$y~.,data=d_train,
+ control=rpart.control(maxdepth=10),method="class")   # run the classificatio
n tree
> plot(ctree, asp=30, main="Business Success")     # Plot ctree # Please chang
e asp to suitable value
> text(ctree, use.n=T, cex = 0.6)                          # Add text   # Plea
se change cex to suitable value
> print(ctree)                                          # Write down the class
ification tree in words
n= 3707

node), split, n, loss, yval, (yprob)
      * denotes terminal node

1) root 3707 410 0 (0.88939844 0.11060156)
  2) nr.employed>=5087.65 3240 207 0 (0.93611111 0.06388889) *
  3) nr.employed< 5087.65 467 203 0 (0.56531049 0.43468951)
    6) pdays>=12.5 358 124 0 (0.65363128 0.34636872) *
    7) pdays< 12.5 109  30 1 (0.27522936 0.72477064) *
>
> prob=predict(ctree, d_train)                          # Prediction on trainin
g data
> col=max.col(prob)                                     # Col is the col. no s.
t. pr is max
> table1 = table(col,d_train[,20])                      # Cross tabulation tabl
e for training dataset new_d
> table1

col    0    1
  1 3267  331
  2   30   79
>
> n<-sum(table1)                                # Number of data
> misclass_d1<-(n-sum(diag(table1)))/n          # Compute d_train misclassificat
ion rate
> misclass_d1
[1] 0.09738333
>
> probD2=predict(ctree,d_test)                          # Prediction on testing
 data
> col2=max.col(probD2)                                  # Col is the col. no s.
t. pr is max
> table2 = table(col2,d_test[,20])                      # Cross tabulation tabl
e for training dataset d_test
> table2

col2   0   1
   1 366  32
   2   5   9
>
> n<-sum(table2)                                # Number of data
> misclass_d2<-(n-sum(diag(table2)))/n          # Compute d_test misclassificati
on rate
> misclass_d2
[1] 0.08980583
>
> end.time <- Sys.time()
> time.taken <- round(end.time - start.time,2)
> time.taken
Time difference of 1.63 secs
```

Appendix V Artificial Neural Network Output (Full Dataset)

```
> start.time <- Sys.time()
>
> library(nnet)                  # Load nnet library
> d<-read.csv("bank-additional-full-PREPROCESSED.csv") # load dataset
> d$duration<-NULL
> names(d)
 [1] "age"          "job"          "marital"      "education"   "default"

 [6] "housing"      "loan"         "contact"      "month"       "day_of_week"

[11] "campaign"     "pdays"        "previous"     "poutcome"    "emp.var.rate"

[16] "cons.price.idx" "cons.conf.idx"  "euribor3m"    "nr.employed"    "y"

>
>
>
> set.seed(74020+82860+74011+71533)   # Use student id to generate sample
> n = max(row(d))
> id = sample(1:n, size = n*9/10)  # generate random row index for d_train
> d_train = d[id,]                        # training dataset
> d_test = d[-id,]                        # testing dataset
> dim(d_train)
[1] 37069    20
> dim(d_test)
[1] 4119   20
>
>
> # ANN
> ann<-function(x,y,size,minsize=1, maxsize=10, maxit=100,linout=F,try=5) {
            # build the improved ann function
+ for(j in minsize:maxsize){
+        ann1<-nnet(y~.,data=x,size=j,maxit=maxit,linout=linout)      # save t
he first result
+          v1<-ann1$value                                                 # s
ave the first value
+
+           for (i in 2:try) {
+                 ann<-nnet(y~.,data=x,size=j,maxit=maxit,linout=linout)   #
i-th trial of ann
+                 if (ann$value<v1) {
 # if the record is better then ever, save it
+                       v1<-ann$value
  # save the best value
+                       ann1<-ann
  # save the best result
+                 }
+           }
+        if (j == minsize){
+                v2<-v1
  # save the first value when j = minsize
+                ann2<-ann1
  # save the first result when j = minsize
+        }
+        if(v1<v2){
+                v2<-v1
```

```
+                 ann2<-ann1
   # return the best result
+         }
+ }
+ ann2
+ }
>
> final_ann <- ann(d_train[,1:19],d_train[,20], minsize = 1, maxsize = 18, lin
out=F)  # Run ANN, selecting best node size from 1 to 18 corresponding to test
ing data
# weights:  55
initial  value 11354.862729
final  value 4187.000000
converged
# weights:  55
initial  value 11725.975126
final  value 4187.000000
converged
# weights:  55
initial  value 10587.261246
final  value 4187.000000
converged
# weights:  55
initial  value 9210.583370
final  value 4187.000000
converged
# weights:  55
initial  value 4692.525626
final  value 4187.000000
converged
# weights:  109
initial  value 13658.856044
final  value 4187.000000
converged
# weights:  109
initial  value 8495.208451
final  value 4187.000000
converged
# weights:  109
initial  value 10350.413047
final  value 4187.000000
converged
# weights:  109
initial  value 10866.046644
final  value 4187.000000
converged
# weights:  109
initial  value 14194.768633
final  value 4187.000000
converged
# weights:  163
initial  value 8671.576150
final  value 4187.000000
converged
# weights:  163
initial  value 7139.087466
final  value 4187.000000
converged
# weights:  163
initial  value 8506.405096
final  value 4187.000000
```

```
converged
# weights:  163
initial  value 9992.516610
final  value 4187.000000
converged
# weights:  163
initial  value 9628.777961
final  value 4187.000000
converged
# weights:  217
initial  value 6472.924948
final  value 4187.000000
converged
# weights:  217
initial  value 6911.788871
final  value 4187.000000
converged
# weights:  217
initial  value 13687.757300
final  value 4187.000000
converged
# weights:  217
initial  value 13988.794786
final  value 4187.000000
converged
# weights:  217
initial  value 14548.348852
final  value 4187.000000
converged
# weights:  271
initial  value 4996.521314
final  value 4187.000000
converged
# weights:  271
initial  value 10695.124148
final  value 4187.000000
converged
# weights:  271
initial  value 12608.503090
final  value 4187.000000
converged
# weights:  271
initial  value 6835.234501
final  value 4187.000000
converged
# weights:  271
initial  value 11478.115855
final  value 4187.000000
converged
# weights:  325
initial  value 6255.353093
final  value 4187.000000
converged
# weights:  325
initial  value 7314.115883
final  value 4187.000000
converged
# weights:  325
initial  value 5768.099953
final  value 4187.000000
converged
```

```
# weights:  325
initial  value 12996.193495
final  value 4187.000000
converged
# weights:  325
initial  value 5509.465411
final  value 4187.000000
converged
# weights:  379
initial  value 10380.366991
final  value 4187.000000
converged
# weights:  379
initial  value 11998.895027
final  value 4187.000000
converged
# weights:  379
initial  value 4198.052693
final  value 4187.000000
converged
# weights:  379
initial  value 12924.247040
final  value 4187.000000
converged
# weights:  379
initial  value 4052.460298
final  value 3714.071974
converged
# weights:  433
initial  value 5725.106886
final  value 4187.000000
converged
# weights:  433
initial  value 3782.559292
final  value 3714.071974
converged
# weights:  433
initial  value 21833.618081
final  value 4187.000000
converged
# weights:  433
initial  value 4368.164128
final  value 4187.000000
converged
# weights:  433
initial  value 5589.812104
final  value 4187.000000
converged
# weights:  487
initial  value 3959.311138
final  value 3327.400129
converged
# weights:  487
initial  value 19037.800474
final  value 4187.000000
converged
# weights:  487
initial  value 18379.525840
final  value 4187.000000
converged
# weights:  487
```

```
initial  value 23797.112534
final  value 4187.000000
converged
# weights:  487
initial  value 15220.238300
final  value 4187.000000
converged
# weights:  541
initial  value 9111.280192
final  value 4187.000000
converged
# weights:  541
initial  value 15614.185237
final  value 4187.000000
converged
# weights:  541
initial  value 9095.113497
final  value 4187.000000
converged
# weights:  541
initial  value 3714.530787
final  value 3714.071974
converged
# weights:  541
initial  value 7267.888745
final  value 4187.000000
converged
# weights:  595
initial  value 8691.623079
final  value 4187.000000
converged
# weights:  595
initial  value 5979.545185
final  value 4187.000000
converged
# weights:  595
initial  value 6244.131695
final  value 4187.000000
converged
# weights:  595
initial  value 13693.603917
final  value 4187.000000
converged
# weights:  595
initial  value 7256.364239
final  value 4187.000000
converged
# weights:  649
initial  value 22683.101156
final  value 4187.000000
converged
# weights:  649
initial  value 3602.716739
iter  10 value 3327.400130
iter  10 value 3327.400129
iter  10 value 3327.400129
final  value 3327.400129
converged
# weights:  649
initial  value 7900.517254
final  value 4187.000000
```

```
converged
# weights:  649
initial  value 4854.776324
final  value 4187.000000
converged
# weights:  649
initial  value 5341.145784
final  value 4187.000000
converged
# weights:  703
initial  value 3751.346124
iter  10 value 3331.840916
final  value 3327.400129
converged
# weights:  703
initial  value 9321.254933
final  value 4187.000000
converged
# weights:  703
initial  value 10212.448268
final  value 4187.000000
converged
# weights:  703
initial  value 3767.796037
final  value 3714.071974
converged
# weights:  703
initial  value 18535.080459
final  value 4187.000000
converged
# weights:  757
initial  value 23943.962536
final  value 4187.000000
converged
# weights:  757
initial  value 4286.941078
final  value 4187.000000
converged
# weights:  757
initial  value 15971.126141
final  value 4187.000000
converged
# weights:  757
initial  value 3647.844444
final  value 3510.605082
converged
# weights:  757
initial  value 8598.783557
final  value 4187.000000
converged
# weights:  811
initial  value 3755.563320
final  value 3714.071974
converged
# weights:  811
initial  value 6719.211817
final  value 4187.000000
converged
# weights:  811
initial  value 5928.975276
final  value 4187.000000
```

```
converged
# weights:  811
initial  value 9551.293287
final  value 4187.000000
converged
# weights:  811
initial  value 4024.779214
final  value 3714.071974
converged
# weights:  865
initial  value 7717.973058
final  value 4187.000000
converged
# weights:  865
initial  value 5429.769097
final  value 4187.000000
converged
# weights:  865
initial  value 8377.984475
final  value 4187.000000
converged
# weights:  865
initial  value 11072.585600
final  value 4187.000000
converged
# weights:  865
initial  value 22831.813695
final  value 4187.000000
converged
# weights:  919
initial  value 7963.102679
final  value 4187.000000
converged
# weights:  919
initial  value 21932.812222
final  value 4187.000000
converged
# weights:  919
initial  value 10871.165902
final  value 4187.000000
converged
# weights:  919
initial  value 5016.895306
final  value 4187.000000
converged
# weights:  919
initial  value 24279.212048
final  value 4187.000000
converged
# weights:  973
initial  value 9226.608587
final  value 4187.000000
converged
# weights:  973
initial  value 17334.182543
final  value 4187.000000
converged
# weights:  973
initial  value 3753.159172
iter  10 value 3332.873845
final  value 3327.400129
```

```
converged
# weights:  973
initial  value 16743.390439
final  value 4187.000000
converged
# weights:  973
initial  value 9477.875035
final  value 4187.000000
converged
> ann_value <- final_ann$value
>
> summary(final_ann)
a 52-18-1 network with 973 weights
options were -
  b->h1  i1->h1  i2->h1  i3->h1  i4->h1  i5->h1  i6->h1  i7->h1  i8->h1  i9->h
1 i10->h1
   0.62   -0.59    0.28    0.51   -0.50    0.59    0.02   -0.58    0.05    0.0
0   -0.36
i11->h1 i12->h1 i13->h1 i14->h1 i15->h1 i16->h1 i17->h1 i18->h1 i19->h1 i20->h
1 i21->h1
  -0.09   -0.17   -0.29   -0.30   -0.21    0.27    0.62    0.12    0.51   -0.6
9   -0.52
i22->h1 i23->h1 i24->h1 i25->h1 i26->h1 i27->h1 i28->h1 i29->h1 i30->h1 i31->h
1 i32->h1
  -0.46   -0.32    0.43   -0.37   -0.04    0.41    0.21   -0.58   -0.09   -0.5
1    0.05
i33->h1 i34->h1 i35->h1 i36->h1 i37->h1 i38->h1 i39->h1 i40->h1 i41->h1 i42->h
1 i43->h1
   0.37   -0.45    0.59   -0.33    0.34   -0.43    0.34   -0.01   -0.05   -0.4
8   -0.15
i44->h1 i45->h1 i46->h1 i47->h1 i48->h1 i49->h1 i50->h1 i51->h1 i52->h1
   0.40    0.40    0.08   -0.42    0.32   -0.52    0.06    0.12   -0.29
  b->h2  i1->h2  i2->h2  i3->h2  i4->h2  i5->h2  i6->h2  i7->h2  i8->h2  i9->h
2 i10->h2
  -0.50    0.50   -0.22    0.44   -0.26    0.54    0.47   -0.66    0.30    0.1
9    0.25
i11->h2 i12->h2 i13->h2 i14->h2 i15->h2 i16->h2 i17->h2 i18->h2 i19->h2 i20->h
2 i21->h2
  -0.27    0.24   -0.37   -0.56    0.68   -0.54    0.25    0.41    0.15    0.4
0   -0.02
i22->h2 i23->h2 i24->h2 i25->h2 i26->h2 i27->h2 i28->h2 i29->h2 i30->h2 i31->h
2 i32->h2
   0.01    0.12   -0.24   -0.40   -0.37   -0.28   -0.22   -0.11   -0.68   -0.4
5   -0.08
i33->h2 i34->h2 i35->h2 i36->h2 i37->h2 i38->h2 i39->h2 i40->h2 i41->h2 i42->h
2 i43->h2
   0.21   -0.41    0.56   -0.39   -0.57   -0.45   -0.47    0.00   -0.67   -0.5
3   -0.47
i44->h2 i45->h2 i46->h2 i47->h2 i48->h2 i49->h2 i50->h2 i51->h2 i52->h2
   0.43    0.41    0.22   -0.10    0.34   -0.63   -0.01    0.63   -0.60
  b->h3  i1->h3  i2->h3  i3->h3  i4->h3  i5->h3  i6->h3  i7->h3  i8->h3  i9->h
3 i10->h3
   0.22    0.09    0.16   -0.17    0.21   -0.03   -0.46   -0.45    0.52   -0.5
4   -0.22
i11->h3 i12->h3 i13->h3 i14->h3 i15->h3 i16->h3 i17->h3 i18->h3 i19->h3 i20->h
3 i21->h3
   0.08    0.34    0.33   -0.37    0.02    0.08    0.69   -0.04    0.47    0.2
8   -0.56
i22->h3 i23->h3 i24->h3 i25->h3 i26->h3 i27->h3 i28->h3 i29->h3 i30->h3 i31->h
3 i32->h3
```

```
   0.10   -0.08    0.05   -0.16   -0.37    0.10   -0.47    0.14    0.08   -0.4
4    0.49
i33->h3 i34->h3 i35->h3 i36->h3 i37->h3 i38->h3 i39->h3 i40->h3 i41->h3 i42->h
3 i43->h3
   0.13    0.66   -0.63    0.04    0.00    0.46    0.21    0.13    0.35   -0.5
2   -0.61
i44->h3 i45->h3 i46->h3 i47->h3 i48->h3 i49->h3 i50->h3 i51->h3 i52->h3
   0.29   -0.11    0.46   -0.67   -0.08    0.63   -0.59   -0.18    0.41
  b->h4  i1->h4  i2->h4  i3->h4  i4->h4  i5->h4  i6->h4  i7->h4  i8->h4  i9->h
4 i10->h4
  -0.52   -0.39   -0.49   -0.64   -0.54    0.63   -0.48   -0.19   -0.11    0.0
7   -0.30
i11->h4 i12->h4 i13->h4 i14->h4 i15->h4 i16->h4 i17->h4 i18->h4 i19->h4 i20->h
4 i21->h4
  -0.31   -0.19    0.34   -0.57   -0.29    0.09    0.00   -0.01    0.36   -0.6
1   -0.54
i22->h4 i23->h4 i24->h4 i25->h4 i26->h4 i27->h4 i28->h4 i29->h4 i30->h4 i31->h
4 i32->h4
  -0.53   -0.32    0.12    0.03   -0.41   -0.21   -0.25    0.20    0.25    0.2
4   -0.27
i33->h4 i34->h4 i35->h4 i36->h4 i37->h4 i38->h4 i39->h4 i40->h4 i41->h4 i42->h
4 i43->h4
   0.46   -0.50    0.23   -0.54   -0.39   -0.31    0.55   -0.02    0.25   -0.3
7   -0.43
i44->h4 i45->h4 i46->h4 i47->h4 i48->h4 i49->h4 i50->h4 i51->h4 i52->h4
  -0.61   -0.01   -0.39    0.15    0.26   -0.05    0.44   -0.26    0.25
  b->h5  i1->h5  i2->h5  i3->h5  i4->h5  i5->h5  i6->h5  i7->h5  i8->h5  i9->h
5 i10->h5
   0.30   -0.34    0.07   -0.38   -0.68    0.35   -0.56   -0.39   -0.54    0.0
8   -0.15
i11->h5 i12->h5 i13->h5 i14->h5 i15->h5 i16->h5 i17->h5 i18->h5 i19->h5 i20->h
5 i21->h5
  -0.11    0.65    0.09    0.12   -0.09   -0.23    0.08   -0.46   -0.51    0.1
5   -0.27
i22->h5 i23->h5 i24->h5 i25->h5 i26->h5 i27->h5 i28->h5 i29->h5 i30->h5 i31->h
5 i32->h5
   0.35   -0.27   -0.06   -0.60   -0.30    0.68   -0.59   -0.52    0.36   -0.1
2   -0.12
i33->h5 i34->h5 i35->h5 i36->h5 i37->h5 i38->h5 i39->h5 i40->h5 i41->h5 i42->h
5 i43->h5
  -0.14   -0.37    0.51   -0.45    0.22    0.29    0.35    0.27    0.37    0.1
9    0.50
i44->h5 i45->h5 i46->h5 i47->h5 i48->h5 i49->h5 i50->h5 i51->h5 i52->h5
  -0.56    0.03   -0.64    0.67   -0.65    0.26   -0.35    0.16    0.08
  b->h6  i1->h6  i2->h6  i3->h6  i4->h6  i5->h6  i6->h6  i7->h6  i8->h6  i9->h
6 i10->h6
   0.59    0.40    0.20   -0.64    0.03    0.39    0.52    0.18    0.52    0.3
3    0.14
i11->h6 i12->h6 i13->h6 i14->h6 i15->h6 i16->h6 i17->h6 i18->h6 i19->h6 i20->h
6 i21->h6
  -0.18    0.28   -0.53   -0.05    0.04    0.40    0.31   -0.35    0.34    0.3
0   -0.22
i22->h6 i23->h6 i24->h6 i25->h6 i26->h6 i27->h6 i28->h6 i29->h6 i30->h6 i31->h
6 i32->h6
  -0.39   -0.31   -0.25    0.11   -0.02    0.18    0.57   -0.29   -0.43    0.4
7    0.07
i33->h6 i34->h6 i35->h6 i36->h6 i37->h6 i38->h6 i39->h6 i40->h6 i41->h6 i42->h
6 i43->h6
  -0.50    0.17    0.48   -0.68   -0.55    0.13   -0.40    0.24   -0.07   -0.1
1   -0.28
i44->h6 i45->h6 i46->h6 i47->h6 i48->h6 i49->h6 i50->h6 i51->h6 i52->h6
```

```
  -0.21   -0.27    0.43    0.04   -0.11    0.48   -0.04    0.55    0.56
  b->h7  i1->h7  i2->h7  i3->h7  i4->h7  i5->h7  i6->h7  i7->h7  i8->h7  i9->h
7 i10->h7
   0.16   -0.10   -0.04    0.08    0.27    0.59   -0.70    0.38   -0.50    0.6
4    0.23
i11->h7 i12->h7 i13->h7 i14->h7 i15->h7 i16->h7 i17->h7 i18->h7 i19->h7 i20->h
7 i21->h7
   0.64   -0.06   -0.49    0.26    0.09    0.70   -0.39   -0.55    0.67   -0.0
5    0.06
i22->h7 i23->h7 i24->h7 i25->h7 i26->h7 i27->h7 i28->h7 i29->h7 i30->h7 i31->h
7 i32->h7
  -0.55   -0.54    0.17    0.38    0.14   -0.64    0.07    0.12   -0.36    0.2
8    0.19
i33->h7 i34->h7 i35->h7 i36->h7 i37->h7 i38->h7 i39->h7 i40->h7 i41->h7 i42->h
7 i43->h7
   0.66    0.09    0.56    0.20    0.05    0.24    0.30    0.15    0.47   -0.5
4    0.53
i44->h7 i45->h7 i46->h7 i47->h7 i48->h7 i49->h7 i50->h7 i51->h7 i52->h7
   0.23   -0.70   -0.19   -0.21   -0.03    0.54   -0.62    0.22    0.10
  b->h8  i1->h8  i2->h8  i3->h8  i4->h8  i5->h8  i6->h8  i7->h8  i8->h8  i9->h
8 i10->h8
   0.35   -0.25    0.07   -0.56   -0.02   -0.59   -0.68   -0.28    0.42    0.2
4   -0.25
i11->h8 i12->h8 i13->h8 i14->h8 i15->h8 i16->h8 i17->h8 i18->h8 i19->h8 i20->h
8 i21->h8
  -0.51    0.27   -0.39    0.41   -0.68   -0.13   -0.14    0.53    0.03    0.3
5    0.59
i22->h8 i23->h8 i24->h8 i25->h8 i26->h8 i27->h8 i28->h8 i29->h8 i30->h8 i31->h
8 i32->h8
  -0.28   -0.34    0.67   -0.07    0.41    0.45   -0.26   -0.50   -0.38   -0.0
9   -0.45
i33->h8 i34->h8 i35->h8 i36->h8 i37->h8 i38->h8 i39->h8 i40->h8 i41->h8 i42->h
8 i43->h8
  -0.16   -0.06   -0.09   -0.43   -0.26    0.17    0.49   -0.41    0.39   -0.4
0   -0.41
i44->h8 i45->h8 i46->h8 i47->h8 i48->h8 i49->h8 i50->h8 i51->h8 i52->h8
  -0.53   -0.09    0.05   -0.61    0.42   -0.18    0.46    0.49    0.58
  b->h9  i1->h9  i2->h9  i3->h9  i4->h9  i5->h9  i6->h9  i7->h9  i8->h9  i9->h
9 i10->h9
   0.10   -0.14   -0.58    0.56   -0.57   -0.35    0.61    0.18    0.21    0.2
7   -0.04
i11->h9 i12->h9 i13->h9 i14->h9 i15->h9 i16->h9 i17->h9 i18->h9 i19->h9 i20->h
9 i21->h9
   0.00    0.21   -0.36    0.08   -0.19    0.21   -0.13    0.64    0.60    0.0
2    0.67
i22->h9 i23->h9 i24->h9 i25->h9 i26->h9 i27->h9 i28->h9 i29->h9 i30->h9 i31->h
9 i32->h9
   0.47   -0.39   -0.12    0.09    0.61    0.62    0.39    0.13    0.24   -0.2
4    0.41
i33->h9 i34->h9 i35->h9 i36->h9 i37->h9 i38->h9 i39->h9 i40->h9 i41->h9 i42->h
9 i43->h9
   0.49    0.00   -0.38    0.43    0.24   -0.62    0.64    0.59   -0.44   -0.3
3    0.26
i44->h9 i45->h9 i46->h9 i47->h9 i48->h9 i49->h9 i50->h9 i51->h9 i52->h9
  -0.56    0.50    0.52   -0.14    0.63   -0.66    0.69   -0.52   -0.30
  b->h10  i1->h10  i2->h10  i3->h10  i4->h10  i5->h10  i6->h10  i7->h10  i8->h
10  i9->h10
   -0.26    -0.58     0.36     0.58    -0.37    -0.30    -0.21     0.24    -0.
38     0.56
i10->h10 i11->h10 i12->h10 i13->h10 i14->h10 i15->h10 i16->h10 i17->h10 i18->h
10 i19->h10
```

```
    0.56    -0.22     0.00    -0.54    -0.58     0.10    -0.01    -0.38     0.
40     0.37
i20->h10 i21->h10 i22->h10 i23->h10 i24->h10 i25->h10 i26->h10 i27->h10 i28->h
10 i29->h10
   -0.62    -0.24     0.02     0.23     0.02    -0.08    -0.57     0.27    -0.
52     0.66
i30->h10 i31->h10 i32->h10 i33->h10 i34->h10 i35->h10 i36->h10 i37->h10 i38->h
10 i39->h10
    0.49    -0.05    -0.59    -0.59    -0.52    -0.51    -0.24    -0.35     0.
39    -0.55
i40->h10 i41->h10 i42->h10 i43->h10 i44->h10 i45->h10 i46->h10 i47->h10 i48->h
10 i49->h10
   -0.17    -0.52    -0.15     0.00    -0.50    -0.62     0.17    -0.66     0.
30    -0.05
i50->h10 i51->h10 i52->h10
   -0.18     0.60    -0.64
  b->h11  i1->h11  i2->h11  i3->h11  i4->h11  i5->h11  i6->h11  i7->h11  i8->h
11  i9->h11
    0.36     0.32     0.54     0.03    -0.64     0.28    -0.08    -0.19     0.
43     0.38
i10->h11 i11->h11 i12->h11 i13->h11 i14->h11 i15->h11 i16->h11 i17->h11 i18->h
11 i19->h11
    0.21     0.54    -0.18     0.26     0.06    -0.09     0.48     0.53     0.
11    -0.05
i20->h11 i21->h11 i22->h11 i23->h11 i24->h11 i25->h11 i26->h11 i27->h11 i28->h
11 i29->h11
   -0.58    -0.57    -0.68     0.05     0.53    -0.56    -0.16     0.23    -0.
67    -0.70
i30->h11 i31->h11 i32->h11 i33->h11 i34->h11 i35->h11 i36->h11 i37->h11 i38->h
11 i39->h11
   -0.24    -0.67    -0.57    -0.16    -0.58     0.24    -0.38    -0.06     0.
59     0.39
i40->h11 i41->h11 i42->h11 i43->h11 i44->h11 i45->h11 i46->h11 i47->h11 i48->h
11 i49->h11
    0.60     0.43     0.48     0.26     0.13    -0.09    -0.45    -0.34    -0.
67    -0.06
i50->h11 i51->h11 i52->h11
    0.35     0.31     0.43
  b->h12  i1->h12  i2->h12  i3->h12  i4->h12  i5->h12  i6->h12  i7->h12  i8->h
12  i9->h12
    0.39    -0.19    -0.48     0.60     0.65    -0.20     0.60     0.27     0.
04     0.40
i10->h12 i11->h12 i12->h12 i13->h12 i14->h12 i15->h12 i16->h12 i17->h12 i18->h
12 i19->h12
   -0.54     0.58    -0.25    -0.33    -0.15     0.43    -0.50     0.13     0.
51     0.53
i20->h12 i21->h12 i22->h12 i23->h12 i24->h12 i25->h12 i26->h12 i27->h12 i28->h
12 i29->h12
   -0.33    -0.53    -0.47    -0.17    -0.15    -0.30    -0.27    -0.29     0.
66    -0.65
i30->h12 i31->h12 i32->h12 i33->h12 i34->h12 i35->h12 i36->h12 i37->h12 i38->h
12 i39->h12
   -0.51     0.01     0.68    -0.14    -0.33     0.11    -0.49    -0.63    -0.
37     0.42
i40->h12 i41->h12 i42->h12 i43->h12 i44->h12 i45->h12 i46->h12 i47->h12 i48->h
12 i49->h12
    0.09     0.64     0.57    -0.59    -0.42    -0.64    -0.61    -0.28     0.
13    -0.63
i50->h12 i51->h12 i52->h12
    0.40     0.58    -0.09
```

```
  b->h13  i1->h13  i2->h13  i3->h13  i4->h13  i5->h13  i6->h13  i7->h13  i8->h
13  i9->h13
   -0.34     0.55     0.47    -0.50     0.51    -0.67    -0.27     0.70     0.
42    -0.19
i10->h13 i11->h13 i12->h13 i13->h13 i14->h13 i15->h13 i16->h13 i17->h13 i18->h
13 i19->h13
    0.34     0.40     0.36    -0.03    -0.70    -0.37     0.05    -0.49     0.
34    -0.47
i20->h13 i21->h13 i22->h13 i23->h13 i24->h13 i25->h13 i26->h13 i27->h13 i28->h
13 i29->h13
    0.42     0.51     0.68    -0.10    -0.69    -0.69     0.56     0.63     0.
60     0.26
i30->h13 i31->h13 i32->h13 i33->h13 i34->h13 i35->h13 i36->h13 i37->h13 i38->h
13 i39->h13
    0.09     0.27    -0.06     0.51    -0.46    -0.29    -0.38     0.25    -0.
45     0.28
i40->h13 i41->h13 i42->h13 i43->h13 i44->h13 i45->h13 i46->h13 i47->h13 i48->h
13 i49->h13
   -0.56     0.50    -0.24    -0.35    -0.44    -0.12    -0.51     0.14     0.
03     0.08
i50->h13 i51->h13 i52->h13
    0.46    -0.19     0.63
  b->h14  i1->h14  i2->h14  i3->h14  i4->h14  i5->h14  i6->h14  i7->h14  i8->h
14  i9->h14
    0.60     0.27     0.13     0.16    -0.55     0.67     0.53    -0.67    -0.
10     0.43
i10->h14 i11->h14 i12->h14 i13->h14 i14->h14 i15->h14 i16->h14 i17->h14 i18->h
14 i19->h14
    0.06     0.00     0.10    -0.47    -0.41     0.48     0.30     0.43     0.
69    -0.05
i20->h14 i21->h14 i22->h14 i23->h14 i24->h14 i25->h14 i26->h14 i27->h14 i28->h
14 i29->h14
   -0.11    -0.12    -0.18     0.54     0.68    -0.27     0.41     0.39    -0.
56     0.05
i30->h14 i31->h14 i32->h14 i33->h14 i34->h14 i35->h14 i36->h14 i37->h14 i38->h
14 i39->h14
   -0.69     0.28    -0.43     0.41     0.49     0.67    -0.52     0.45    -0.
56     0.37
i40->h14 i41->h14 i42->h14 i43->h14 i44->h14 i45->h14 i46->h14 i47->h14 i48->h
14 i49->h14
   -0.35    -0.53     0.33    -0.48    -0.10     0.15     0.69    -0.52     0.
49     0.44
i50->h14 i51->h14 i52->h14
    0.37     0.28     0.63
  b->h15  i1->h15  i2->h15  i3->h15  i4->h15  i5->h15  i6->h15  i7->h15  i8->h
15  i9->h15
    0.08    -0.10    -0.24     0.41     0.41    -0.27    -0.37    -0.04     0.
10     0.38
i10->h15 i11->h15 i12->h15 i13->h15 i14->h15 i15->h15 i16->h15 i17->h15 i18->h
15 i19->h15
   -0.21    -0.68    -0.45    -0.26     0.52    -0.40     0.44     0.30     0.
08     0.15
i20->h15 i21->h15 i22->h15 i23->h15 i24->h15 i25->h15 i26->h15 i27->h15 i28->h
15 i29->h15
    0.28     0.28    -0.43     0.40     0.32    -0.68    -0.44    -0.44     0.
30    -0.69
i30->h15 i31->h15 i32->h15 i33->h15 i34->h15 i35->h15 i36->h15 i37->h15 i38->h
15 i39->h15
    0.70     0.60    -0.01    -0.65    -0.22     0.13     0.44    -0.20    -0.
61    -0.46
```

```
i40->h15 i41->h15 i42->h15 i43->h15 i44->h15 i45->h15 i46->h15 i47->h15 i48->h
15 i49->h15
    0.44    -0.48    -0.39     0.03     0.60    -0.28     0.67    -0.25    -0.
69    -0.25
i50->h15 i51->h15 i52->h15
   -0.69    -0.54     0.69
  b->h16  i1->h16  i2->h16  i3->h16  i4->h16  i5->h16  i6->h16  i7->h16  i8->h
16  i9->h16
    0.44     0.17     0.13    -0.39    -0.20     0.30    -0.19     0.34     0.
56    -0.23
i10->h16 i11->h16 i12->h16 i13->h16 i14->h16 i15->h16 i16->h16 i17->h16 i18->h
16 i19->h16
    0.09     0.52     0.58     0.01     0.14     0.30     0.47     0.28     0.
40    -0.17
i20->h16 i21->h16 i22->h16 i23->h16 i24->h16 i25->h16 i26->h16 i27->h16 i28->h
16 i29->h16
    0.68     0.38    -0.65     0.16     0.65    -0.65     0.47    -0.58     0.
27    -0.58
i30->h16 i31->h16 i32->h16 i33->h16 i34->h16 i35->h16 i36->h16 i37->h16 i38->h
16 i39->h16
   -0.51    -0.51    -0.18    -0.01    -0.32     0.20     0.26     0.44     0.
33     0.09
i40->h16 i41->h16 i42->h16 i43->h16 i44->h16 i45->h16 i46->h16 i47->h16 i48->h
16 i49->h16
   -0.07     0.47    -0.61     0.22     0.04     0.64    -0.56    -0.34     0.
42    -0.07
i50->h16 i51->h16 i52->h16
    0.02    -0.29    -0.23
  b->h17  i1->h17  i2->h17  i3->h17  i4->h17  i5->h17  i6->h17  i7->h17  i8->h
17  i9->h17
    0.45     0.04    -0.61     0.17     0.59     0.33     0.50    -0.50     0.
21    -0.39
i10->h17 i11->h17 i12->h17 i13->h17 i14->h17 i15->h17 i16->h17 i17->h17 i18->h
17 i19->h17
   -0.24     0.27    -0.59    -0.59     0.01     0.50     0.22    -0.51     0.
29     0.11
i20->h17 i21->h17 i22->h17 i23->h17 i24->h17 i25->h17 i26->h17 i27->h17 i28->h
17 i29->h17
    0.60     0.50    -0.31    -0.39     0.52     0.12     0.23    -0.23    -0.
13     0.40
i30->h17 i31->h17 i32->h17 i33->h17 i34->h17 i35->h17 i36->h17 i37->h17 i38->h
17 i39->h17
   -0.69     0.19    -0.21     0.11     0.01    -0.45    -0.05    -0.33    -0.
39     0.36
i40->h17 i41->h17 i42->h17 i43->h17 i44->h17 i45->h17 i46->h17 i47->h17 i48->h
17 i49->h17
    0.26    -0.32    -0.41    -0.55     0.01    -0.24    -0.46    -0.05    -0.
62     0.38
i50->h17 i51->h17 i52->h17
   -0.39    -0.45    -0.35
  b->h18  i1->h18  i2->h18  i3->h18  i4->h18  i5->h18  i6->h18  i7->h18  i8->h
18  i9->h18
    0.53     0.31    -0.47    -0.17    -0.14     0.09     0.59    -0.34    -0.
45     0.30
i10->h18 i11->h18 i12->h18 i13->h18 i14->h18 i15->h18 i16->h18 i17->h18 i18->h
18 i19->h18
   -0.49    -0.55     0.55    -0.39    -0.47    -0.54    -0.66    -0.06     0.
63    -0.01
i20->h18 i21->h18 i22->h18 i23->h18 i24->h18 i25->h18 i26->h18 i27->h18 i28->h
18 i29->h18
```

```
   -0.04    -0.44     0.33    -0.23     0.08     0.13    -0.28     0.05     0.
25    -0.59
i30->h18 i31->h18 i32->h18 i33->h18 i34->h18 i35->h18 i36->h18 i37->h18 i38->h
18 i39->h18
   -0.35     0.13    -0.11     0.56     0.40    -0.44     0.53     0.29    -0.
22     0.41
i40->h18 i41->h18 i42->h18 i43->h18 i44->h18 i45->h18 i46->h18 i47->h18 i48->h
18 i49->h18
   -0.33     0.57    -0.03    -0.30     0.23     0.57    -0.59     0.08     0.
38    -0.22
i50->h18 i51->h18 i52->h18
    0.63    -0.64     0.49
  b->o  h1->o  h2->o  h3->o  h4->o  h5->o  h6->o  h7->o  h8->o  h9->o h10->o h
11->o h12->o
 -0.56  -0.54  -0.44  -0.39  -0.69   2.83   0.47  -0.71   0.08   0.38   0.04
-0.13   0.51
h13->o h14->o h15->o h16->o h17->o h18->o
  0.08  -0.32   0.07  -0.68  -0.07  -0.17
> # Show summary
of final_ann
> pred<-1*(final_ann$fit>1/2)
> table1<-table(pred,d_train$y)
> table1

pred     0     1
   0 32381  3318
   1   501   869
> n<-sum(table1) # Number of dat
a
> misclass_train<-(n-sum(diag(table1)))/n # Compute d_tra
in misclassification rate
> misclass_train # Show
cross tabulate table for training dataset
[1] 0.1030241
>
> pr<-predict(final_ann, d_test)
> pred<-1*(pr>1/2)
> table2<-table(pred, d_test$y)
> table2

pred    0    1
   0 3619  355
   1   47   98
> n<-sum(table2) # Number of dat
a
> misclass_test<-(n-sum(diag(table2)))/n # Compute d_test
 misclassification rate
> misclass_test
[1] 0.0975965
>
> end.time <- Sys.time()
> time.taken <- round(end.time - start.time,2)
> time.taken
Time difference of 2.99 mins
```

Appendix VI Artificial Neural Network Output (10% Dataset)

```
> start.time <- Sys.time()
>
> library(nnet)                  # Load nnet library
> d<-read.csv("bank-additional-PREPROCESSED.csv") # load dataset
> d$duration<-NULL
> names(d)
 [1] "age"            "job"            "marital"        "education"
 [5] "default"        "housing"        "loan"           "contact"
 [9] "month"          "day_of_week"    "campaign"       "pdays"
[13] "previous"       "poutcome"       "emp.var.rate"   "cons.price.idx"
[17] "cons.conf.idx"  "euribor3m"      "nr.employed"    "y"
>
>
>
> set.seed(74020+82860+74011+71533)   # Use student id to generate sample
> n = max(row(d))
> id = sample(1:n, size = n*9/10)  # generate random row index for d_train
> d_train = d[id,]                       # training dataset
> d_test = d[-id,]                       # testing dataset
> dim(d_train)
[1] 3707   20
> dim(d_test)
[1] 412  20
>
>
> # ANN
> ann<-function(x,y,size,minsize=1, maxsize=10, maxit=100,linout=F,try=5) {
            # build the improved ann function
+ for(j in minsize:maxsize){
```

```
+        ann1<-nnet(y~.,data=x,size=j,maxit=maxit,linout=linout)      # save t
he first result
+          v1<-ann1$value                                                 # s
ave the first value
+
+           for (i in 2:try) {
+                 ann<-nnet(y~.,data=x,size=j,maxit=maxit,linout=linout)   #
i-th trial of ann
+                 if (ann$value<v1) {
 # if the record is better then ever, save it
+                       v1<-ann$value
  # save the best value
+                       ann1<-ann
  # save the best result
+                 }
+           }
+        if (j == minsize){
+                 v2<-v1
  # save the first value when j = minsize
+                 ann2<-ann1
  # save the first result when j = minsize
+        }
+        if(v1<v2){
+                 v2<-v1
+                 ann2<-ann1
   # return the best result
+        }
+ }
+ ann2
+ }
>
> final_ann <- ann(d_train[,1:19],d_train[,20], minsize = 1, maxsize = 18, lin
out=F)  # Run ANN, selecting best node size from 1 to 18 corresponding to test
ing data
# weights:  55
initial  value 613.584542
final  value 410.000000
converged
# weights:  55
initial  value 489.626205
final  value 410.000000
converged
# weights:  55
initial  value 1042.497041
final  value 410.000000
converged
# weights:  55
initial  value 824.549113
final  value 410.000000
converged
# weights:  55
initial  value 589.246206
final  value 410.000000
converged
# weights:  109
initial  value 474.023432
final  value 410.000000
converged
# weights:  109
initial  value 1458.476305
final  value 410.000000
```

```
converged
# weights:  109
initial  value 642.068043
final  value 410.000000
converged
# weights:  109
initial  value 469.829662
final  value 410.000000
converged
# weights:  109
initial  value 789.639459
final  value 410.000000
converged
# weights:  163
initial  value 971.921012
final  value 410.000000
converged
# weights:  163
initial  value 684.520726
final  value 410.000000
converged
# weights:  163
initial  value 741.203239
final  value 410.000000
converged
# weights:  163
initial  value 1820.992166
final  value 410.000000
converged
# weights:  163
initial  value 643.972149
final  value 410.000000
converged
# weights:  217
initial  value 877.690547
final  value 410.000000
converged
# weights:  217
initial  value 1410.123333
final  value 410.000000
converged
# weights:  217
initial  value 1286.847945
final  value 410.000000
converged
# weights:  217
initial  value 902.003549
final  value 410.000000
converged
# weights:  217
initial  value 1254.461728
final  value 410.000000
converged
# weights:  271
initial  value 720.862145
final  value 410.000000
converged
# weights:  271
initial  value 1262.557581
final  value 410.000000
converged
```

```
# weights:  271
initial  value 1389.526143
final  value 410.000000
converged
# weights:  271
initial  value 418.898679
final  value 410.000000
converged
# weights:  271
initial  value 581.694848
final  value 410.000000
converged
# weights:  325
initial  value 553.229780
final  value 410.000000
converged
# weights:  325
initial  value 2095.144044
final  value 410.000000
converged
# weights:  325
initial  value 931.786285
final  value 410.000000
converged
# weights:  325
initial  value 567.228537
final  value 410.000000
converged
# weights:  325
initial  value 984.363025
final  value 410.000000
converged
# weights:  379
initial  value 1055.033249
final  value 410.000000
converged
# weights:  379
initial  value 1133.853842
final  value 410.000000
converged
# weights:  379
initial  value 2284.441514
final  value 410.000000
converged
# weights:  379
initial  value 1012.082555
final  value 410.000000
converged
# weights:  379
initial  value 2028.624986
final  value 410.000000
converged
# weights:  433
initial  value 705.900984
final  value 410.000000
converged
# weights:  433
initial  value 467.039357
final  value 410.000000
converged
# weights:  433
```

```
initial  value 969.691773
final  value 410.000000
converged
# weights:  433
initial  value 1415.894001
final  value 410.000000
converged
# weights:  433
initial  value 947.852144
final  value 410.000000
converged
# weights:  487
initial  value 1496.675455
final  value 410.000000
converged
# weights:  487
initial  value 1359.954615
final  value 410.000000
converged
# weights:  487
initial  value 366.783602
final  value 364.653359
converged
# weights:  487
initial  value 2362.023496
final  value 410.000000
converged
# weights:  487
initial  value 916.083092
final  value 410.000000
converged
# weights:  541
initial  value 513.485487
final  value 410.000000
converged
# weights:  541
initial  value 1238.989065
final  value 410.000000
converged
# weights:  541
initial  value 798.341262
final  value 410.000000
converged
# weights:  541
initial  value 1905.797423
final  value 410.000000
converged
# weights:  541
initial  value 647.941840
final  value 410.000000
converged
# weights:  595
initial  value 374.608851
final  value 364.653359
converged
# weights:  595
initial  value 1447.433634
final  value 410.000000
converged
# weights:  595
initial  value 1323.009479
```

```
final  value 410.000000
converged
# weights:  595
initial  value 999.997758
final  value 410.000000
converged
# weights:  595
initial  value 537.899885
final  value 410.000000
converged
# weights:  649
initial  value 441.047416
final  value 410.000000
converged
# weights:  649
initial  value 399.842096
final  value 364.653359
converged
# weights:  649
initial  value 1553.432488
final  value 410.000000
converged
# weights:  649
initial  value 1082.405288
final  value 410.000000
converged
# weights:  649
initial  value 2072.388645
final  value 410.000000
converged
# weights:  703
initial  value 1050.495311
final  value 410.000000
converged
# weights:  703
initial  value 414.735400
final  value 410.000000
converged
# weights:  703
initial  value 778.852956
final  value 410.000000
converged
# weights:  703
initial  value 384.015034
final  value 364.653359
converged
# weights:  703
initial  value 542.689808
final  value 410.000000
converged
# weights:  757
initial  value 402.498035
final  value 323.138639
converged
# weights:  757
initial  value 551.926925
final  value 410.000000
converged
# weights:  757
initial  value 628.694208
final  value 410.000000
```

```
converged
# weights:  757
initial  value 651.174605
final  value 410.000000
converged
# weights:  757
initial  value 383.097172
final  value 364.653359
converged
# weights:  811
initial  value 452.839461
final  value 410.000000
converged
# weights:  811
initial  value 1106.927506
final  value 410.000000
converged
# weights:  811
initial  value 636.526312
final  value 410.000000
converged
# weights:  811
initial  value 595.122542
final  value 410.000000
converged
# weights:  811
initial  value 762.082286
final  value 410.000000
converged
# weights:  865
initial  value 769.047175
final  value 410.000000
converged
# weights:  865
initial  value 404.794954
final  value 364.653359
converged
# weights:  865
initial  value 1776.889764
final  value 410.000000
converged
# weights:  865
initial  value 870.746030
final  value 410.000000
converged
# weights:  865
initial  value 603.697863
final  value 410.000000
converged
# weights:  919
initial  value 1056.629490
final  value 410.000000
converged
# weights:  919
initial  value 1695.644923
final  value 410.000000
converged
# weights:  919
initial  value 844.770970
final  value 410.000000
converged
```

```
# weights:  919
initial  value 1772.281368
final  value 410.000000
converged
# weights:  919
initial  value 1676.797414
final  value 410.000000
converged
# weights:  973
initial  value 368.707223
final  value 341.455540
converged
# weights:  973
initial  value 1672.202740
final  value 410.000000
converged
# weights:  973
initial  value 462.694098
final  value 410.000000
converged
# weights:  973
initial  value 1223.382520
final  value 410.000000
converged
# weights:  973
initial  value 949.028300
final  value 410.000000
converged
> ann_value <- final_ann$value
>
> summary(final_ann)
a 52-14-1 network with 757 weights
options were -
  b->h1  i1->h1  i2->h1  i3->h1  i4->h1  i5->h1  i6->h1  i7->h1  i8->h1
   0.04    0.01    0.15    0.11   -0.14   -0.57    0.53    0.64    0.52
 i9->h1 i10->h1 i11->h1 i12->h1 i13->h1 i14->h1 i15->h1 i16->h1 i17->h1
  -0.44    0.25    0.35   -0.62    0.52    0.15   -0.05   -0.01    0.15
i18->h1 i19->h1 i20->h1 i21->h1 i22->h1 i23->h1 i24->h1 i25->h1 i26->h1
  -0.31    0.48    0.67    0.64   -0.68   -0.07    0.04    0.47   -0.36
i27->h1 i28->h1 i29->h1 i30->h1 i31->h1 i32->h1 i33->h1 i34->h1 i35->h1
   0.37   -0.30    0.59    0.46    0.37    0.06    0.03    0.53    0.54
i36->h1 i37->h1 i38->h1 i39->h1 i40->h1 i41->h1 i42->h1 i43->h1 i44->h1
  -0.08    0.45    0.09    0.50   -0.38   -0.50   -0.38    0.62    0.53
i45->h1 i46->h1 i47->h1 i48->h1 i49->h1 i50->h1 i51->h1 i52->h1
   0.54   -0.61    0.03    0.67   -0.45   -0.69   -0.43    0.53
  b->h2  i1->h2  i2->h2  i3->h2  i4->h2  i5->h2  i6->h2  i7->h2  i8->h2
   0.35    0.17    0.01    0.24   -0.19   -0.33   -0.10   -0.61    0.04
 i9->h2 i10->h2 i11->h2 i12->h2 i13->h2 i14->h2 i15->h2 i16->h2 i17->h2
   0.53   -0.02   -0.15   -0.31   -0.47   -0.55   -0.11    0.08   -0.23
i18->h2 i19->h2 i20->h2 i21->h2 i22->h2 i23->h2 i24->h2 i25->h2 i26->h2
  -0.14    0.43   -0.03    0.35   -0.17   -0.61   -0.19    0.03    0.55
i27->h2 i28->h2 i29->h2 i30->h2 i31->h2 i32->h2 i33->h2 i34->h2 i35->h2
   0.69    0.66    0.21    0.54   -0.11   -0.14   -0.46   -0.08    0.37
i36->h2 i37->h2 i38->h2 i39->h2 i40->h2 i41->h2 i42->h2 i43->h2 i44->h2
  -0.62   -0.04    0.70   -0.17    0.56    0.16   -0.16   -0.41    0.51
i45->h2 i46->h2 i47->h2 i48->h2 i49->h2 i50->h2 i51->h2 i52->h2
   0.13    0.42   -0.10   -0.67   -0.11   -0.61   -0.55   -0.32
  b->h3  i1->h3  i2->h3  i3->h3  i4->h3  i5->h3  i6->h3  i7->h3  i8->h3
  -0.34    0.25    0.43    0.41   -0.28    0.30   -0.31    0.12   -0.30
 i9->h3 i10->h3 i11->h3 i12->h3 i13->h3 i14->h3 i15->h3 i16->h3 i17->h3
  -0.28    0.67    0.57    0.28    0.22    0.46   -0.35    0.69   -0.57
```

```
i18->h3 i19->h3 i20->h3 i21->h3 i22->h3 i23->h3 i24->h3 i25->h3 i26->h3
   0.50   -0.39    0.42    0.43   -0.22   -0.04   -0.47   -0.66    0.60
i27->h3 i28->h3 i29->h3 i30->h3 i31->h3 i32->h3 i33->h3 i34->h3 i35->h3
   0.42   -0.05    0.08    0.45   -0.02   -0.40   -0.29   -0.43   -0.11
i36->h3 i37->h3 i38->h3 i39->h3 i40->h3 i41->h3 i42->h3 i43->h3 i44->h3
   0.16    0.57   -0.60   -0.52   -0.17    0.26    0.00   -0.24   -0.41
i45->h3 i46->h3 i47->h3 i48->h3 i49->h3 i50->h3 i51->h3 i52->h3
   0.50    0.04   -0.20   -0.03   -0.16    0.52   -0.30    0.35
  b->h4  i1->h4  i2->h4  i3->h4  i4->h4  i5->h4  i6->h4  i7->h4  i8->h4
  -0.42    0.53    0.04    0.21   -0.59   -0.41   -0.51   -0.25   -0.58
 i9->h4 i10->h4 i11->h4 i12->h4 i13->h4 i14->h4 i15->h4 i16->h4 i17->h4
  -0.62    0.44    0.30   -0.25    0.02   -0.40    0.58   -0.65   -0.10
i18->h4 i19->h4 i20->h4 i21->h4 i22->h4 i23->h4 i24->h4 i25->h4 i26->h4
  -0.35    0.44   -0.59   -0.59    0.66    0.01   -0.41    0.47   -0.64
i27->h4 i28->h4 i29->h4 i30->h4 i31->h4 i32->h4 i33->h4 i34->h4 i35->h4
  -0.36    0.42   -0.39    0.65   -0.44    0.68   -0.69   -0.36    0.46
i36->h4 i37->h4 i38->h4 i39->h4 i40->h4 i41->h4 i42->h4 i43->h4 i44->h4
   0.24    0.06    0.66    0.35    0.18   -0.05   -0.31    0.26    0.66
i45->h4 i46->h4 i47->h4 i48->h4 i49->h4 i50->h4 i51->h4 i52->h4
  -0.26   -0.43    0.03   -0.18    0.64   -0.14   -0.42   -0.09
  b->h5  i1->h5  i2->h5  i3->h5  i4->h5  i5->h5  i6->h5  i7->h5  i8->h5
  -0.05   -0.08   -0.15    0.62   -0.37   -0.05    0.50    0.02   -0.25
 i9->h5 i10->h5 i11->h5 i12->h5 i13->h5 i14->h5 i15->h5 i16->h5 i17->h5
  -0.68    0.46    0.17   -0.55    0.33   -0.63    0.09   -0.47    0.11
i18->h5 i19->h5 i20->h5 i21->h5 i22->h5 i23->h5 i24->h5 i25->h5 i26->h5
   0.59    0.05   -0.24    0.04   -0.42    0.16   -0.18    0.00    0.56
i27->h5 i28->h5 i29->h5 i30->h5 i31->h5 i32->h5 i33->h5 i34->h5 i35->h5
  -0.70    0.63    0.01    0.43    0.15   -0.54    0.13   -0.16   -0.52
i36->h5 i37->h5 i38->h5 i39->h5 i40->h5 i41->h5 i42->h5 i43->h5 i44->h5
  -0.54    0.03   -0.48    0.60    0.24    0.01   -0.68   -0.43    0.31
i45->h5 i46->h5 i47->h5 i48->h5 i49->h5 i50->h5 i51->h5 i52->h5
  -0.26    0.35    0.07   -0.22   -0.36   -0.26   -0.24    0.15
  b->h6  i1->h6  i2->h6  i3->h6  i4->h6  i5->h6  i6->h6  i7->h6  i8->h6
  -0.55   -0.07    0.01   -0.25    0.64    0.48    0.14   -0.60   -0.27
 i9->h6 i10->h6 i11->h6 i12->h6 i13->h6 i14->h6 i15->h6 i16->h6 i17->h6
   0.06   -0.08   -0.17   -0.44   -0.29    0.69   -0.30   -0.10    0.32
i18->h6 i19->h6 i20->h6 i21->h6 i22->h6 i23->h6 i24->h6 i25->h6 i26->h6
  -0.08   -0.11   -0.58    0.41    0.70    0.65   -0.16   -0.21   -0.08
i27->h6 i28->h6 i29->h6 i30->h6 i31->h6 i32->h6 i33->h6 i34->h6 i35->h6
  -0.39   -0.42    0.58   -0.01    0.43   -0.23    0.08   -0.33    0.32
i36->h6 i37->h6 i38->h6 i39->h6 i40->h6 i41->h6 i42->h6 i43->h6 i44->h6
  -0.12   -0.07    0.25    0.27   -0.67   -0.07    0.21    0.35    0.34
i45->h6 i46->h6 i47->h6 i48->h6 i49->h6 i50->h6 i51->h6 i52->h6
  -0.14   -0.54    0.14    0.15    0.26    0.68    0.18   -1.22
  b->h7  i1->h7  i2->h7  i3->h7  i4->h7  i5->h7  i6->h7  i7->h7  i8->h7
  -0.22   -0.42    0.64   -0.57   -0.16   -0.02   -0.28    0.58    0.00
 i9->h7 i10->h7 i11->h7 i12->h7 i13->h7 i14->h7 i15->h7 i16->h7 i17->h7
  -0.61   -0.63    0.69    0.27   -0.42    0.16   -0.23    0.63   -0.48
i18->h7 i19->h7 i20->h7 i21->h7 i22->h7 i23->h7 i24->h7 i25->h7 i26->h7
  -0.39    0.64   -0.50    0.43    0.69    0.35    0.43   -0.26    0.37
i27->h7 i28->h7 i29->h7 i30->h7 i31->h7 i32->h7 i33->h7 i34->h7 i35->h7
   0.53   -0.02   -0.10    0.65   -0.15    0.06    0.69   -0.45    0.04
i36->h7 i37->h7 i38->h7 i39->h7 i40->h7 i41->h7 i42->h7 i43->h7 i44->h7
   0.01    0.39   -0.62   -0.13    0.21    0.17    0.41    0.07   -0.54
i45->h7 i46->h7 i47->h7 i48->h7 i49->h7 i50->h7 i51->h7 i52->h7
  -0.55   -0.55    0.13    0.24   -0.27   -0.01    0.10   -0.10
  b->h8  i1->h8  i2->h8  i3->h8  i4->h8  i5->h8  i6->h8  i7->h8  i8->h8
   0.29    0.39    0.56    0.68   -0.32   -0.33    0.54   -0.08    0.08
 i9->h8 i10->h8 i11->h8 i12->h8 i13->h8 i14->h8 i15->h8 i16->h8 i17->h8
   0.32    0.09   -0.41    0.17    0.23   -0.06   -0.55    0.64   -0.58
i18->h8 i19->h8 i20->h8 i21->h8 i22->h8 i23->h8 i24->h8 i25->h8 i26->h8
```

```
   0.57    0.50    0.22   -0.27    0.43   -0.16   -0.01   -0.38    0.47
i27->h8 i28->h8 i29->h8 i30->h8 i31->h8 i32->h8 i33->h8 i34->h8 i35->h8
  -0.58    0.01    0.52    0.10   -0.29   -0.22   -0.05   -0.64   -0.42
i36->h8 i37->h8 i38->h8 i39->h8 i40->h8 i41->h8 i42->h8 i43->h8 i44->h8
   0.12   -0.04    0.30   -0.41   -0.36   -0.37    0.69   -0.36   -0.44
i45->h8 i46->h8 i47->h8 i48->h8 i49->h8 i50->h8 i51->h8 i52->h8
   0.66    0.04    0.38    0.04   -0.29    0.36    0.32   -0.60
  b->h9  i1->h9  i2->h9  i3->h9  i4->h9  i5->h9  i6->h9  i7->h9  i8->h9
  -0.16   -0.67    0.52   -0.39   -0.18    0.22    0.27   -0.27    0.08
 i9->h9 i10->h9 i11->h9 i12->h9 i13->h9 i14->h9 i15->h9 i16->h9 i17->h9
   0.54   -0.57    0.14   -0.40   -0.32   -0.35    0.07    0.10    0.09
i18->h9 i19->h9 i20->h9 i21->h9 i22->h9 i23->h9 i24->h9 i25->h9 i26->h9
  -0.43   -0.44   -0.47   -0.32   -0.09   -0.18   -0.10    0.66    0.48
i27->h9 i28->h9 i29->h9 i30->h9 i31->h9 i32->h9 i33->h9 i34->h9 i35->h9
  -0.15   -0.44    0.66   -0.30    0.14   -0.02   -0.68   -0.32   -0.30
i36->h9 i37->h9 i38->h9 i39->h9 i40->h9 i41->h9 i42->h9 i43->h9 i44->h9
  -0.28   -0.65   -0.02    0.08   -0.66   -0.11    0.19   -0.64    0.44
i45->h9 i46->h9 i47->h9 i48->h9 i49->h9 i50->h9 i51->h9 i52->h9
   0.09   -0.34    0.15    0.35   -0.09   -0.24   -0.45   -0.05
  b->h10  i1->h10  i2->h10  i3->h10  i4->h10  i5->h10  i6->h10  i7->h10
    0.14     0.58    -0.32    -0.69    -0.39     0.29    -0.21     0.37
 i8->h10  i9->h10 i10->h10 i11->h10 i12->h10 i13->h10 i14->h10 i15->h10
    0.06    -0.07    -0.08    -0.47     0.03     0.47     0.33    -0.40
i16->h10 i17->h10 i18->h10 i19->h10 i20->h10 i21->h10 i22->h10 i23->h10
   -0.09     0.64     0.66     0.66    -0.69     0.09    -0.63     0.21
i24->h10 i25->h10 i26->h10 i27->h10 i28->h10 i29->h10 i30->h10 i31->h10
    0.13    -0.19     0.35     0.24     0.31     0.29    -0.08     0.25
i32->h10 i33->h10 i34->h10 i35->h10 i36->h10 i37->h10 i38->h10 i39->h10
   -0.53    -0.51     0.10     0.53     0.10    -0.01     0.20     0.06
i40->h10 i41->h10 i42->h10 i43->h10 i44->h10 i45->h10 i46->h10 i47->h10
   -0.33     0.39    -0.19     0.67    -0.66     0.67    -0.19     0.04
i48->h10 i49->h10 i50->h10 i51->h10 i52->h10
    0.50    -0.33     0.38    -0.58    -0.21
  b->h11  i1->h11  i2->h11  i3->h11  i4->h11  i5->h11  i6->h11  i7->h11
   -0.62    -0.41     0.27    -0.02     0.21    -0.51    -0.15     0.15
 i8->h11  i9->h11 i10->h11 i11->h11 i12->h11 i13->h11 i14->h11 i15->h11
    0.59     0.23     0.15    -0.28    -0.24     0.47     0.05     0.69
i16->h11 i17->h11 i18->h11 i19->h11 i20->h11 i21->h11 i22->h11 i23->h11
   -0.67     0.65     0.18     0.67     0.27     0.22    -0.56    -0.40
i24->h11 i25->h11 i26->h11 i27->h11 i28->h11 i29->h11 i30->h11 i31->h11
    0.36    -0.07     0.24     0.60    -0.06     0.49     0.69    -0.34
i32->h11 i33->h11 i34->h11 i35->h11 i36->h11 i37->h11 i38->h11 i39->h11
    0.62     0.42    -0.55    -0.02    -0.51    -0.59    -0.05    -0.30
i40->h11 i41->h11 i42->h11 i43->h11 i44->h11 i45->h11 i46->h11 i47->h11
   -0.03     0.14    -0.34    -0.63     0.28     0.32    -0.22     0.27
i48->h11 i49->h11 i50->h11 i51->h11 i52->h11
   -0.40    -0.42    -0.62     0.28     0.44
  b->h12  i1->h12  i2->h12  i3->h12  i4->h12  i5->h12  i6->h12  i7->h12
    0.54    -0.62     0.31    -0.13     0.23     0.35    -0.46    -0.19
 i8->h12  i9->h12 i10->h12 i11->h12 i12->h12 i13->h12 i14->h12 i15->h12
    0.63     0.17    -0.06    -0.26    -0.10     0.47     0.26    -0.38
i16->h12 i17->h12 i18->h12 i19->h12 i20->h12 i21->h12 i22->h12 i23->h12
    0.04     0.21    -0.19    -0.67    -0.03    -0.33    -0.25     0.35
i24->h12 i25->h12 i26->h12 i27->h12 i28->h12 i29->h12 i30->h12 i31->h12
    0.45    -0.55     0.59    -0.11    -0.03    -0.32     0.52    -0.44
i32->h12 i33->h12 i34->h12 i35->h12 i36->h12 i37->h12 i38->h12 i39->h12
   -0.03     0.26    -0.33    -0.11     0.60    -0.35     0.15     0.28
i40->h12 i41->h12 i42->h12 i43->h12 i44->h12 i45->h12 i46->h12 i47->h12
   -0.54     0.27    -0.58    -0.03     0.22    -0.10    -0.28     0.45
i48->h12 i49->h12 i50->h12 i51->h12 i52->h12
    0.22     0.11    -0.30     0.48     0.04
```

```
  b->h13  i1->h13  i2->h13  i3->h13  i4->h13  i5->h13  i6->h13  i7->h13
    0.15    -0.04    -0.63    -0.04     0.37    -0.22    -0.60    -0.64
 i8->h13  i9->h13 i10->h13 i11->h13 i12->h13 i13->h13 i14->h13 i15->h13
   -0.58    -0.46    -0.45     0.42    -0.05     0.58     0.28     0.01
i16->h13 i17->h13 i18->h13 i19->h13 i20->h13 i21->h13 i22->h13 i23->h13
    0.06     0.69     0.08    -0.22     0.54    -0.49    -0.25     0.48
i24->h13 i25->h13 i26->h13 i27->h13 i28->h13 i29->h13 i30->h13 i31->h13
    0.03     0.09    -0.21     0.29     0.04     0.65    -0.19    -0.69
i32->h13 i33->h13 i34->h13 i35->h13 i36->h13 i37->h13 i38->h13 i39->h13
   -0.62     0.53     0.30    -0.20     0.36     0.23    -0.68    -0.57
i40->h13 i41->h13 i42->h13 i43->h13 i44->h13 i45->h13 i46->h13 i47->h13
   -0.20     0.53     0.25    -0.52     0.35     0.44    -0.15     0.54
i48->h13 i49->h13 i50->h13 i51->h13 i52->h13
    0.17    -0.38    -0.43    -0.64    -0.24
  b->h14  i1->h14  i2->h14  i3->h14  i4->h14  i5->h14  i6->h14  i7->h14
   -0.49     0.20     0.20    -0.59     0.56    -0.25    -0.55    -0.11
 i8->h14  i9->h14 i10->h14 i11->h14 i12->h14 i13->h14 i14->h14 i15->h14
    0.37     0.26    -0.58     0.15     0.10     0.67     0.43     0.34
i16->h14 i17->h14 i18->h14 i19->h14 i20->h14 i21->h14 i22->h14 i23->h14
   -0.06     0.23     0.33     0.58     0.64    -0.41     0.04    -0.05
i24->h14 i25->h14 i26->h14 i27->h14 i28->h14 i29->h14 i30->h14 i31->h14
   -0.53     0.18     0.16    -0.42    -0.55    -0.26     0.48     0.17
i32->h14 i33->h14 i34->h14 i35->h14 i36->h14 i37->h14 i38->h14 i39->h14
   -0.61     0.03     0.66    -0.19     0.21     0.31    -0.61     0.27
i40->h14 i41->h14 i42->h14 i43->h14 i44->h14 i45->h14 i46->h14 i47->h14
   -0.08    -0.60    -0.22    -0.04    -0.07     0.15     0.52     0.49
i48->h14 i49->h14 i50->h14 i51->h14 i52->h14
   -0.13     0.23    -0.47     0.34    -0.08
  b->o  h1->o  h2->o  h3->o  h4->o  h5->o  h6->o  h7->o  h8->o  h9->o
  0.61   0.00   0.65  -0.10  -1.18  -0.14  -0.53   0.11  -0.29  -1.72
h10->o h11->o h12->o h13->o h14->o
 -0.57  -0.11   0.32   0.31  -0.62
> # Show summary
of final_ann
> pred<-1*(final_ann$fit>1/2)
> table1<-table(pred,d_train$y)
> table1

pred    0    1
   0 3246  319
   1   51   91
> n<-sum(table1)                                    # Number of dat
a
> misclass_train<-(n-sum(diag(table1)))/n           # Compute d_tra
in misclassification rate
> misclass_train                                            # Show
cross tabulate table for training dataset
[1] 0.09981117
>
> pr<-predict(final_ann, d_test)
> pred<-1*(pr>1/2)
> table2<-table(pred, d_test$y)
> table2

pred   0   1
   0 362  32
   1   9   9
> n<-sum(table2)                                    # Number of dat
a
> misclass_test<-(n-sum(diag(table2)))/n           # Compute d_test
 misclassification rate
```

```
> misclass_test
[1] 0.09951456
>
> end.time <- Sys.time()
> time.taken <- round(end.time - start.time,2)
> time.taken
Time difference of 20.8 secs
```